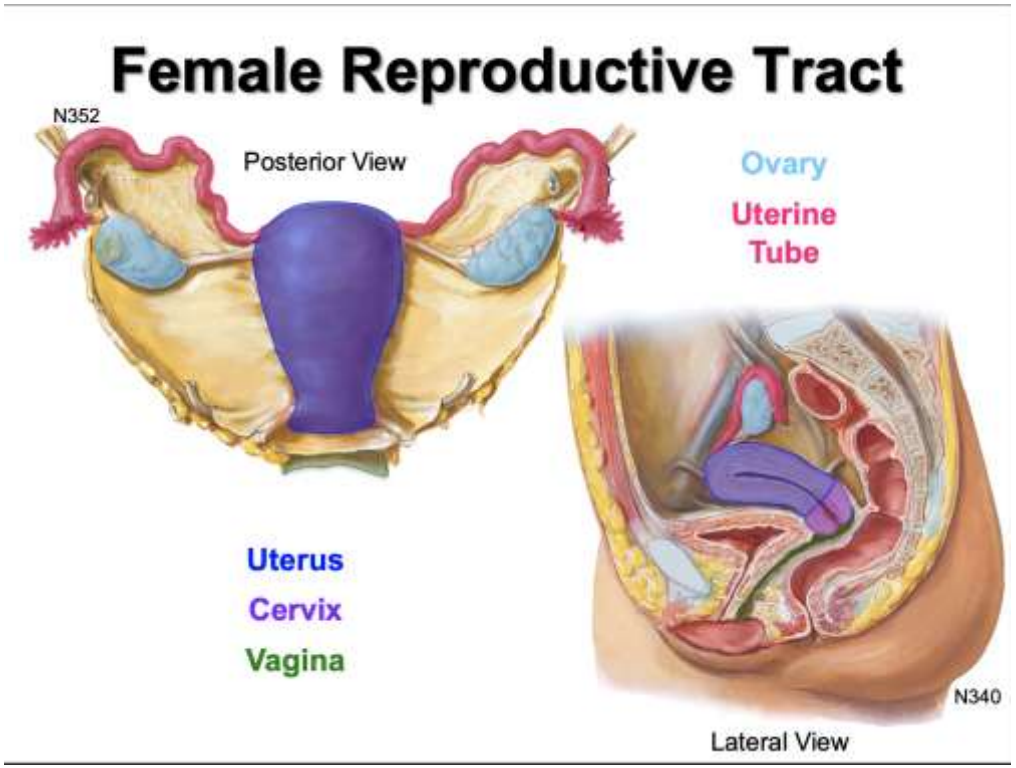
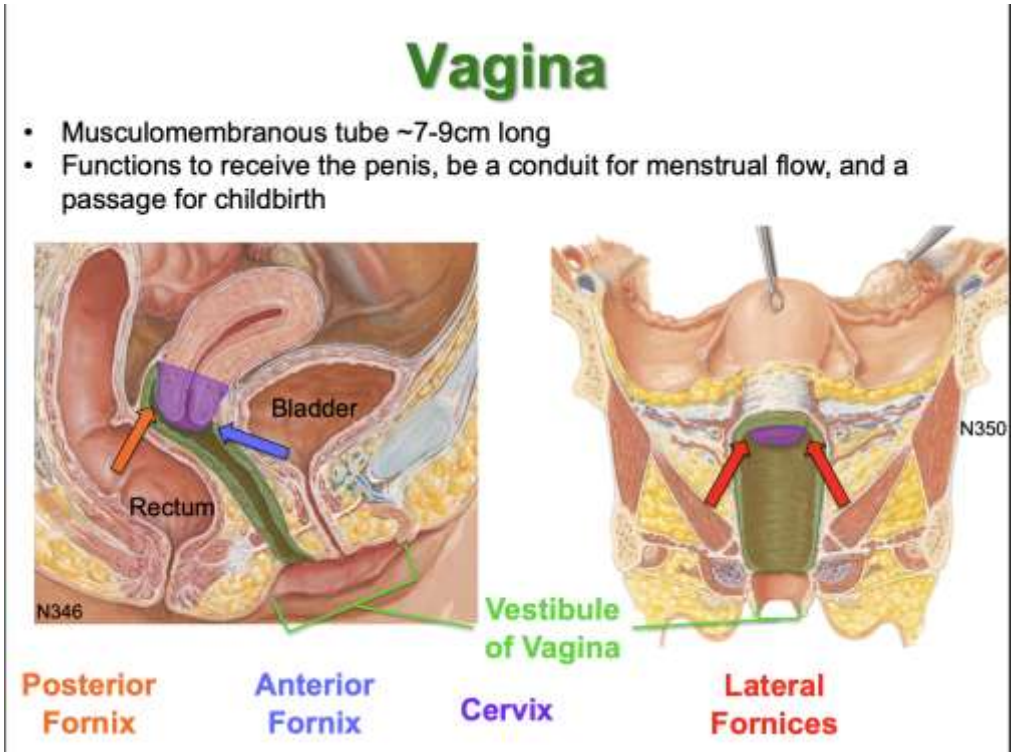
Week #3: [Female Reproductive Tract]

1. Organs of the Female Reproductive Tract
2. Structures of Support
3. Pelvic Floor
4. Nerve Supply
5. Vascular Supply & Lymphatic Drainage
6. Nerve Supply
7. Review of Perineum from P1
8. Urogenital Triangle: Male & Female External Genitalia
9. Anal Triangle
10. Functional Sexual Anatomy

Organs of the Female Reproductive Tract

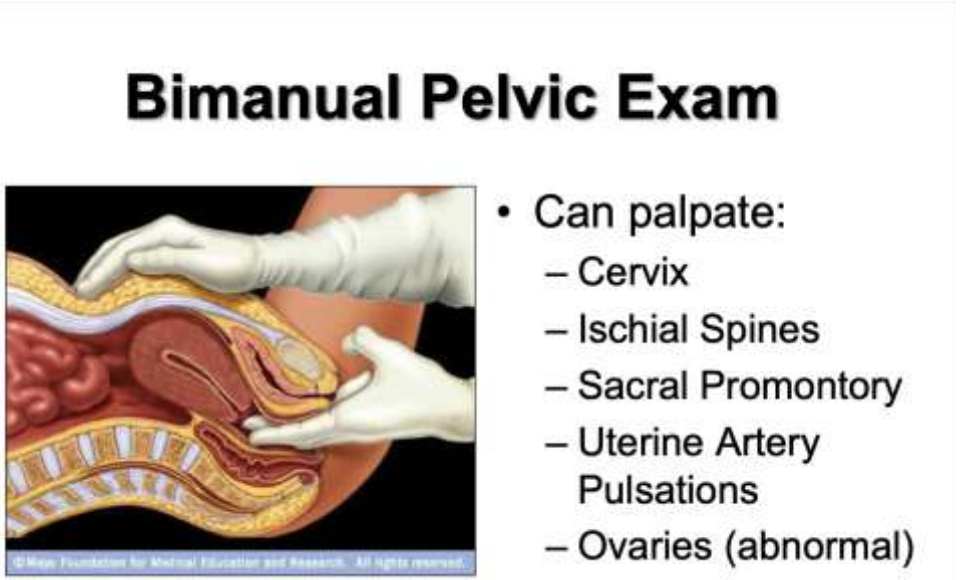




Vagina: musculomembranous tube (7-9cm long) extending from the cervix to the vaginal orifice, which opens into the vestibule of the vagina (between the labia minora).

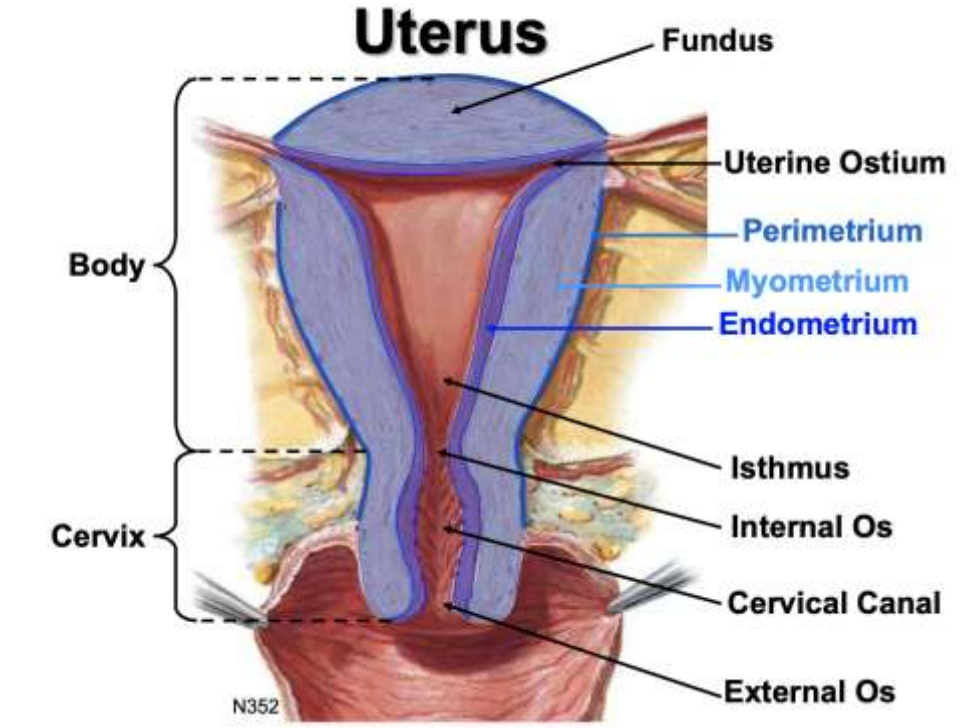
a. Lies between the urethra (anteriorly) and rectum (posteriorly)
b. Capable of remarkable distension during childbirth

c. The **uterine cervix** protrudes into the vagina superiorly, creating a surrounding crevice, the **vaginal fornix (anterior, posterior, and lateral parts)**



Clinical Note: The relatively thin, distensible walls and central location of the vagina enables palpation of the:

a. Cervix
b. Ischial spines
c. Sacral promontory
d. Pulsation of the uterine arteries
e. Irregularities of the ovaries (such as cysts) via **manual pelvic examination.**



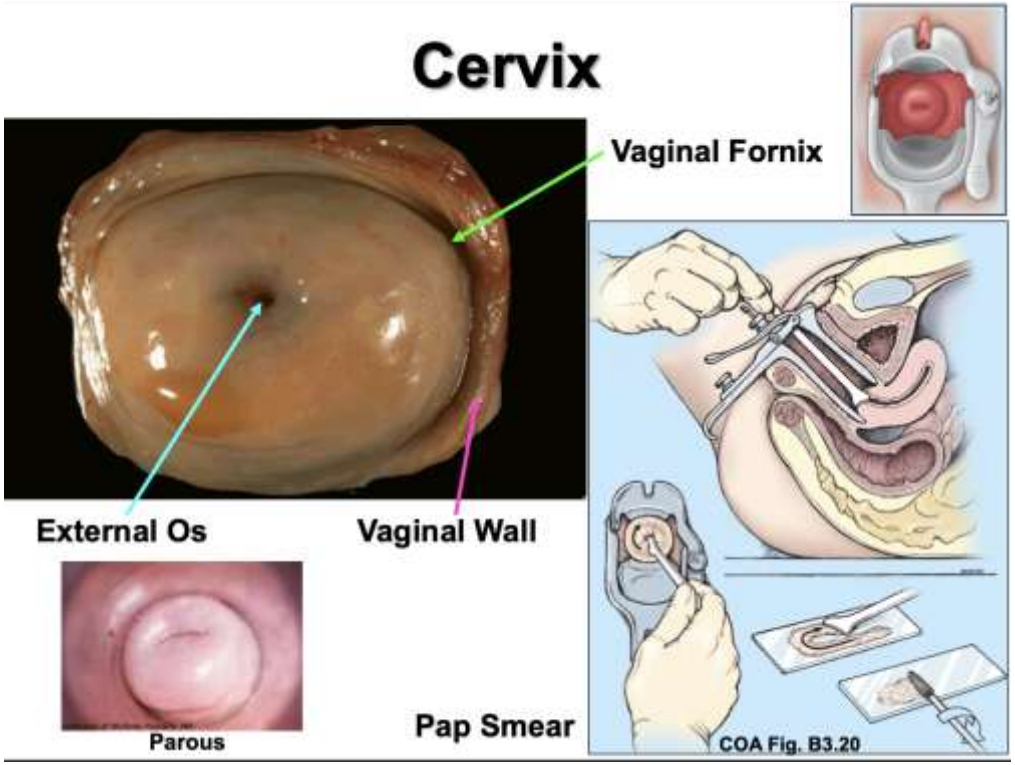
Uterus: thick-walled, hollow muscular organ, where the embryo/fetus develops

1. Parts of the uterus:

a. Uterine body (upper 2/3)
   i. **Fundus** – rounded part superior to the uterine tubes
   ii. **Body** – wide, superior part between the layers of the broad ligament
   iii. **Isthmus** – constricted segment superior to the cervix

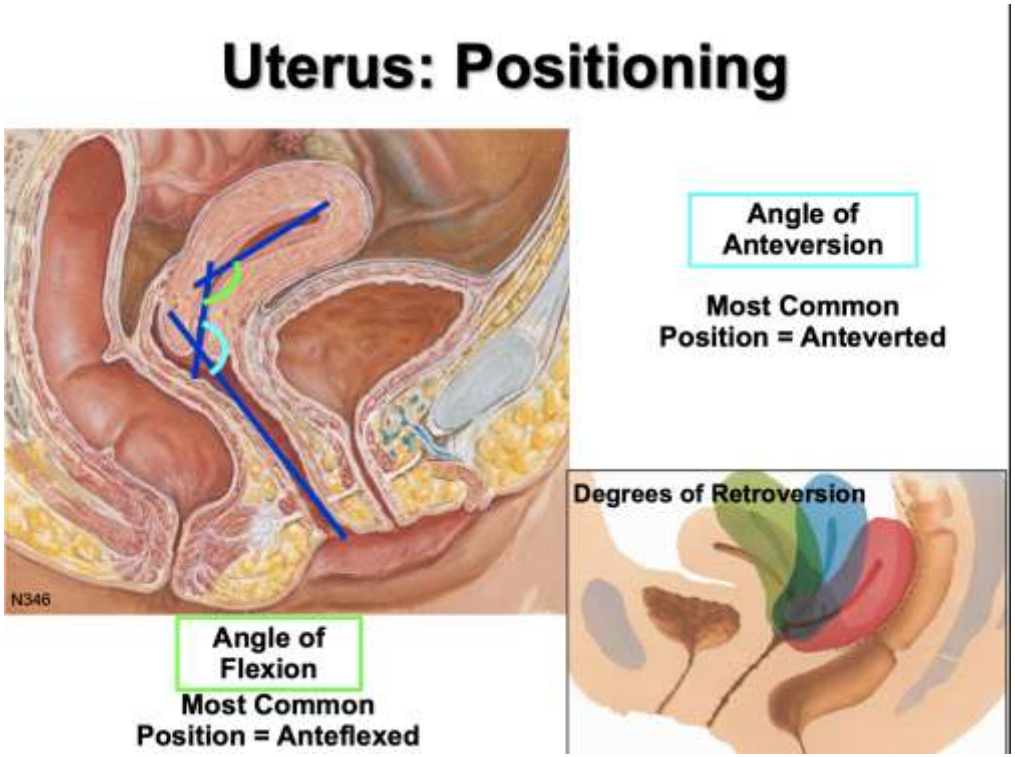
The **wall of the body of the uterus** consists of three layers:

a. **Perimetrium** – outer serous coat
b. **Myometrium** – middle coat of smooth muscle, which contains the neurovasculature. Hormones stimulate contraction (cramping) of this layer during childbirth or during menses
c. **Endometrium** – inner mucous coat, the inner surface of which is shed during menstruation. If conception occurs, the blastocyst implants in this layer



Cervix (lower 1/3) – thicker portion, inferior part protrudes into the upper vagina at the external os (opening to vagina)

Clinical Note: Mortality rates for cervical cancer have declined due to accessibility of cervical exams involving direct visualization of the cervix and Pap smears (cells scraped from the cervix are examined microscopically for abnormal changes). Human papilloma virus (HPV) infections increase the incidence of cervical cancer. A new vaccine has been produced that can drastically reduce the incidence of HPV infections.

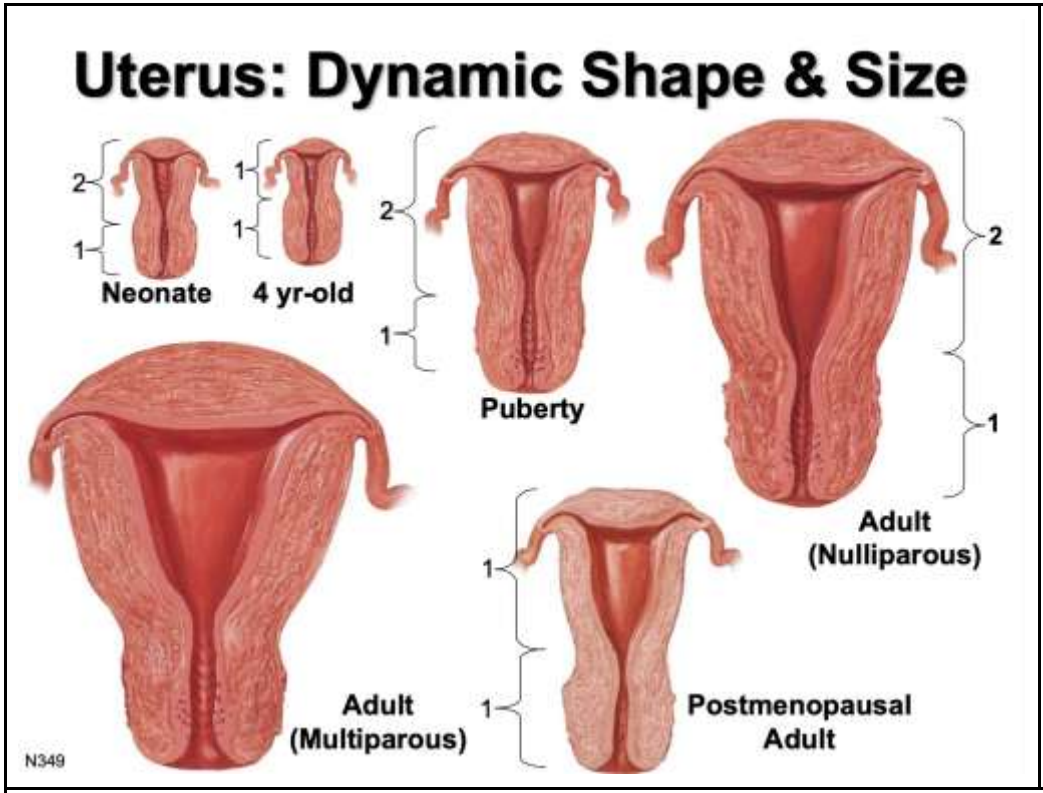


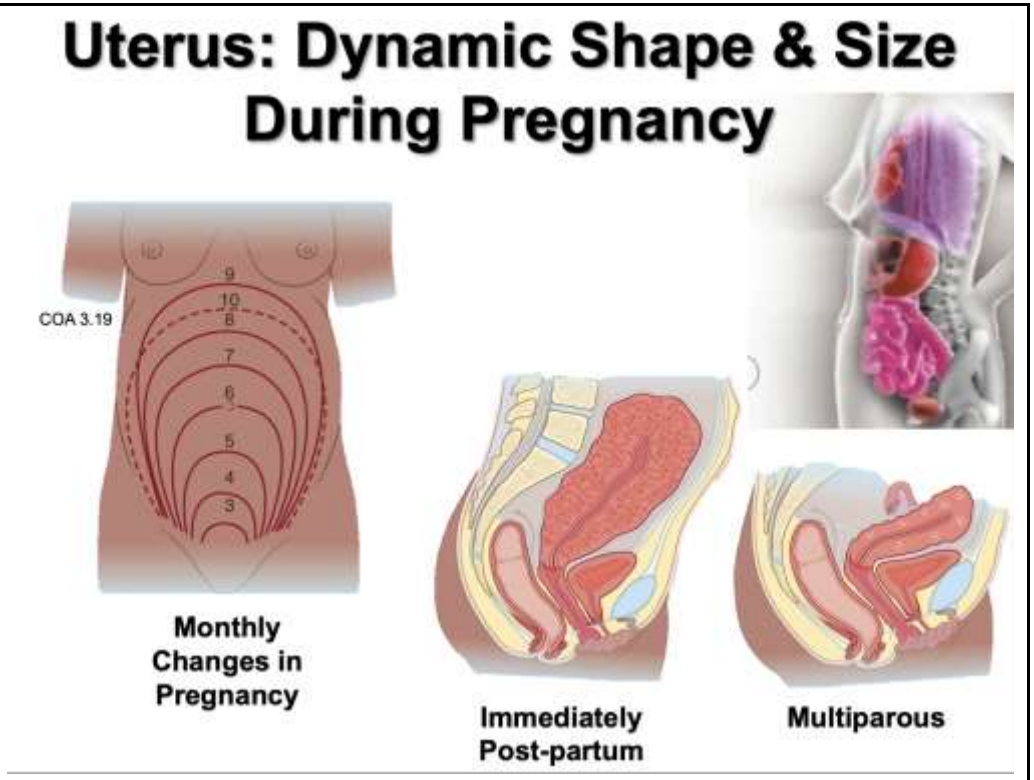
Normal positioning:

a. **Anteverted** – the uterine body is tipped anteriorly, relative to the axis of the vagina (reduced or reversed angle = retroverted)

b. **Anteflexed** – the uterine body is flexed or bent anteriorly relative to the cervix, creating the angle of flexion (straight or 'unbent' or flexed posteriorly = retroflexed)

i. Consequently, the body of the uterus lies in a horizontal plane and rests on top of the empty bladder; with bladder filling, the uterus becomes temporarily retroverted





Dynamic shape and size depending on age or functional status:

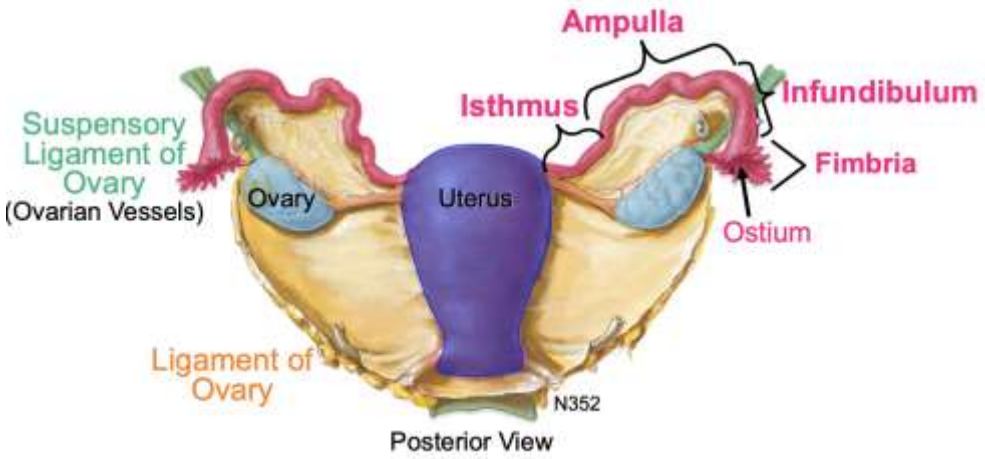
a. Typical appearance of the uterus (post-pubertal, premenopausal nulliparous uterus) – **uterine body:cervix ratio** = **2:1**
b. ***Neonatal uterus*** – under the influence of maternal hormones, thus adult proportions **2:1**
c. ***Childhood uterus*** – in absence of hormones, assumes immature proportion until puberty **1:1**
d. ***Gravid uterus*** (during pregnancy) – becomes very thin walled and extends out of the pelvis, ultimately to the level of the costal margin; rapidly shrinks post- partum
e. ***Multiparous*** (after several pregnancies) – large; doesn't shrink back to its pre- partum size **3:1**
f. ***Postmenopausal uterus*** – in the absence of hormones, resumes immature proportions **1:1**
    i. Size and disposition are determined by bimanual palpation
    ii. Menstrual cycle can affect the appearance of the uterus relative to size and internal consistency

Uterine Tubes & Ovaries

Adnexa of the Uterus

Uterine Tubes:
- Conduct the oocyte/developing embryo from the peritoneal cavity to the uterine cavity

Ovaries:
- Develop oocytes
- Produce reproductive hormones



Uterine (Fallopian) Tubes: tubes extending laterally from the uterus and open into the peritoneal cavity near the ovaries; conduct the oocyte from the peritoneal cavity to the uterine cavity

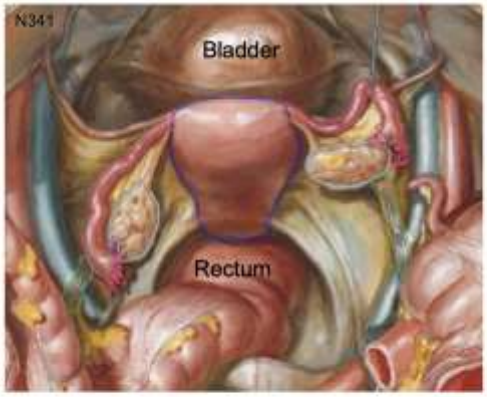
1. Divided into four parts:

a. **Isthmus** – thick-walled, narrowest part of the tube that enters the uterine horn
b. **Ampulla** – widest and longest part of the tube. Fertilization usually occurs here
c. **Infundibulum** – funnel-shaped distal end that opens into the peritoneal cavity. Fimbriae at the end of the infundibulum spread over the medial surface of the ovary to catch the released oocyte
d. **Uterine part** – short segment passing through the uterine wall

Adnexa = anything lateral attachment

Female Reproductive Tract in Situ

Superior View
Anterior

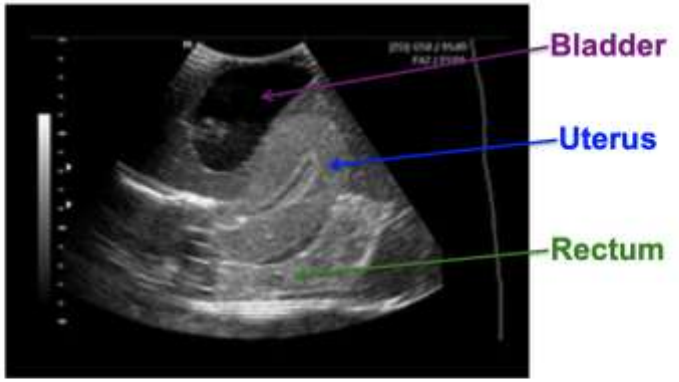


Posterior





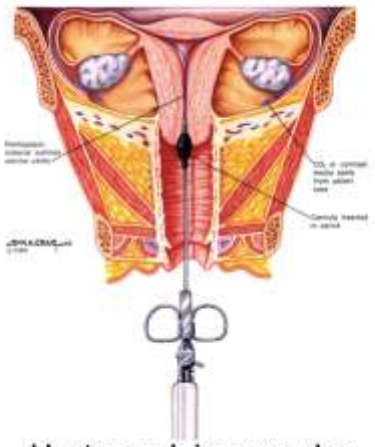
Visualization of Female Reproductive Tract



Functional Visualization of the Uterine Tubes



Cornual obstruction with dye in uterus only



Hysterosalpingography

Clinical Note: A major cause of infertility in women is blockage of the uterine tubes, which may occur due to scarring from disease or inflammation. Patency of the uterine tubes may be determined by injecting radiopaque material or CO_2 into the uterus. The die will pass through the uterus and uterine tubes enabling visualization of these structures by a radiographic procedure known as a ***hysterosalpingography (HSG)***. Patent uterine tubes allow the contrast media to pass through the uterine tubes into the peritoneal cavity.

Clinical Note: Ligation of the uterine tube (**tubal ligation**) is a surgical method of birth control. Oocytes are still discharged from the ovaries and enter the tubes, but degenerate and are soon absorbed.

Ovaries: Ovulation



1 Mature Follicles & a few Immature Follicles



Egg Rupture from Ovary

Ovaries: almond-shaped gonads where the oocytes develop; produce reproductive hormones

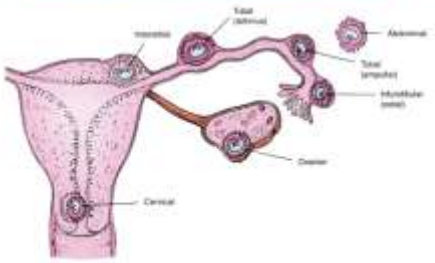
a. **Clinical Note:** Abnormal cysts can form inside the ovaries. Rupture of these can lead to life-threatening hemorrhage. Other types of cysts can develop into ovarian cancer that can become widely disseminated throughout the peritoneal cavity.

Black circles = developing follicles

Ectopic Pregnancy



- Fertilized egg implants outside the uterus
- Most common site – uterine tube
- Rare cases may occur:
  - Ovary
  - Cervix
  - Peritoneal Cavity
- If undetected may cause severe abdominopelvic hemorrhage



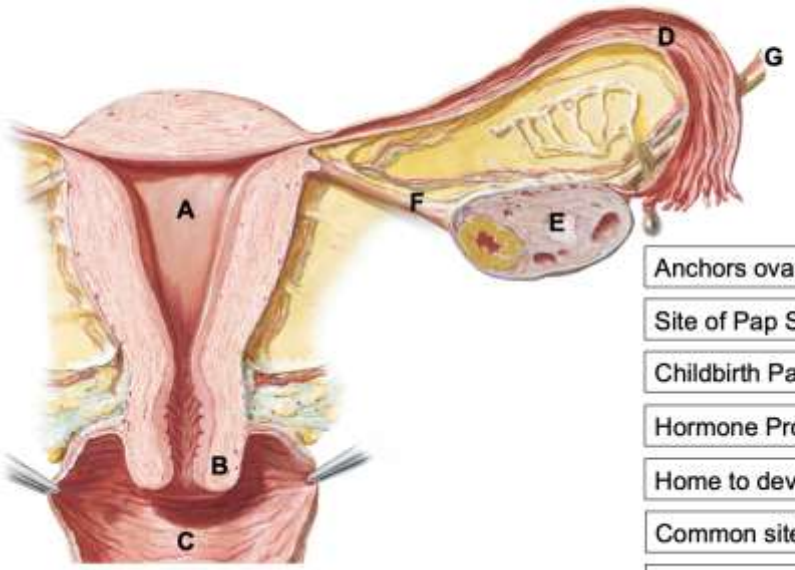


Clinical Note: An **ectopic pregnancy** occurs when a fertilized egg implants outside of the uterus; it is highly unlikely for the baby to survive.

a. The most common site for abnormal implantation is the uterine tube (**ectopic tubal pregnancy**), but may occur in the ovary, cervix, or peritoneal cavity. If not diagnosed early, ectopic tubal pregnancies may rupture the uterine tube causing severe hemorrhaging into the abdominopelvic cavity, threatening the mother's life.

Knowledge Check

Match the correct function to the labelled structures



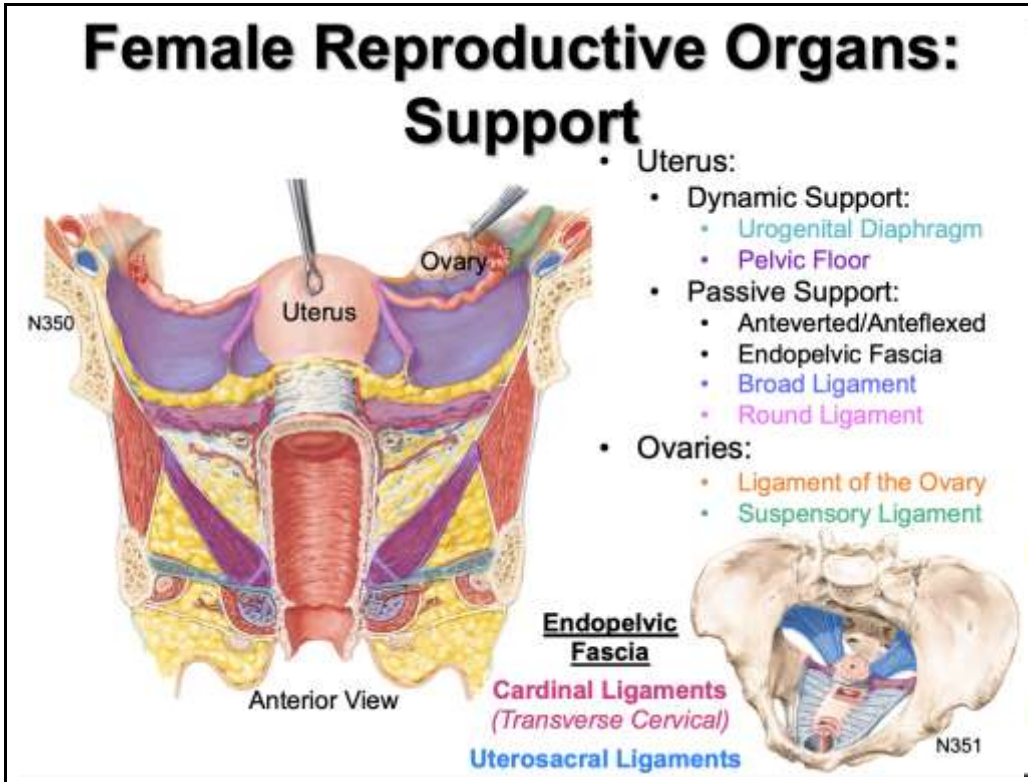
- Anchors ovary to uterus
- Site of Pap Smears
- Childbirth Passage
- Hormone Producer
- Home to developing fetus
- Common site of fertilization
- Carries ovarian vessels

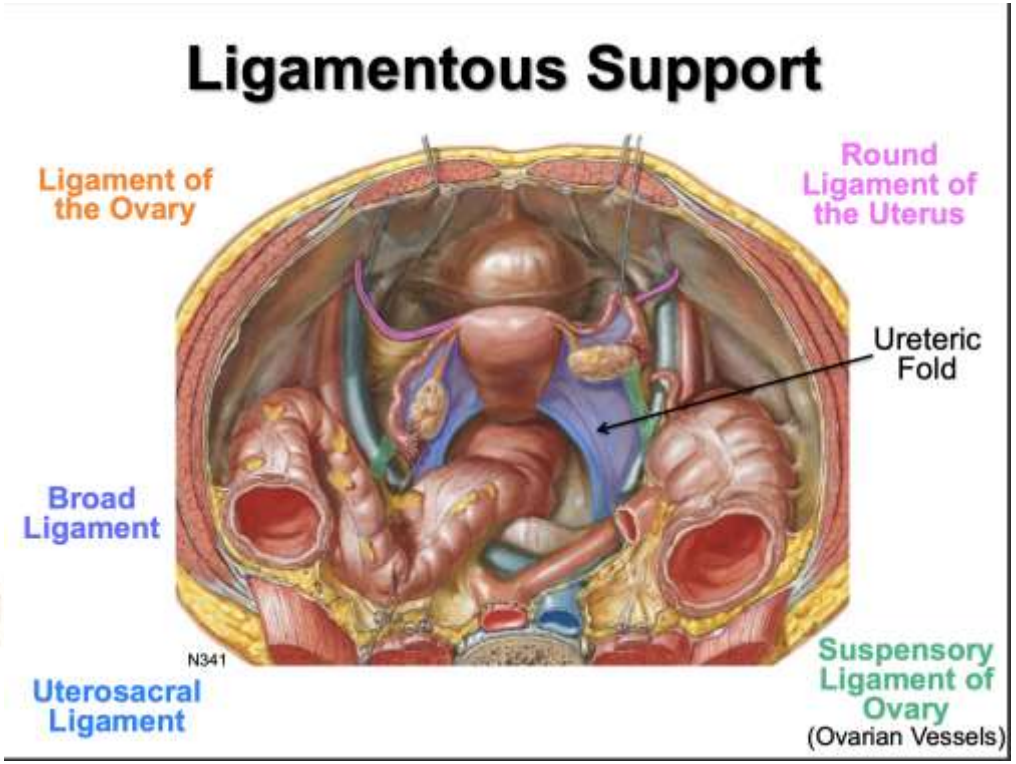
Home to Developing Fetus	A
Site of Pap Smears	B
Childbirth Passage	C
Common Site of Fertilization	D
Hormone Producer	E
Anchors Ovary to Uterus	F
Carries Ovarian Vessels	G

- The female reproductive organs sit sub-peritoneal in the pelvis
- The uterus, cervix, and vagina sit centrally within the pelvis
- The adnexal structures, the uterine tubes and ovaries, sit lateral and posterior to the uterus within the pelvis

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STRUCTURES OF SUPPORT

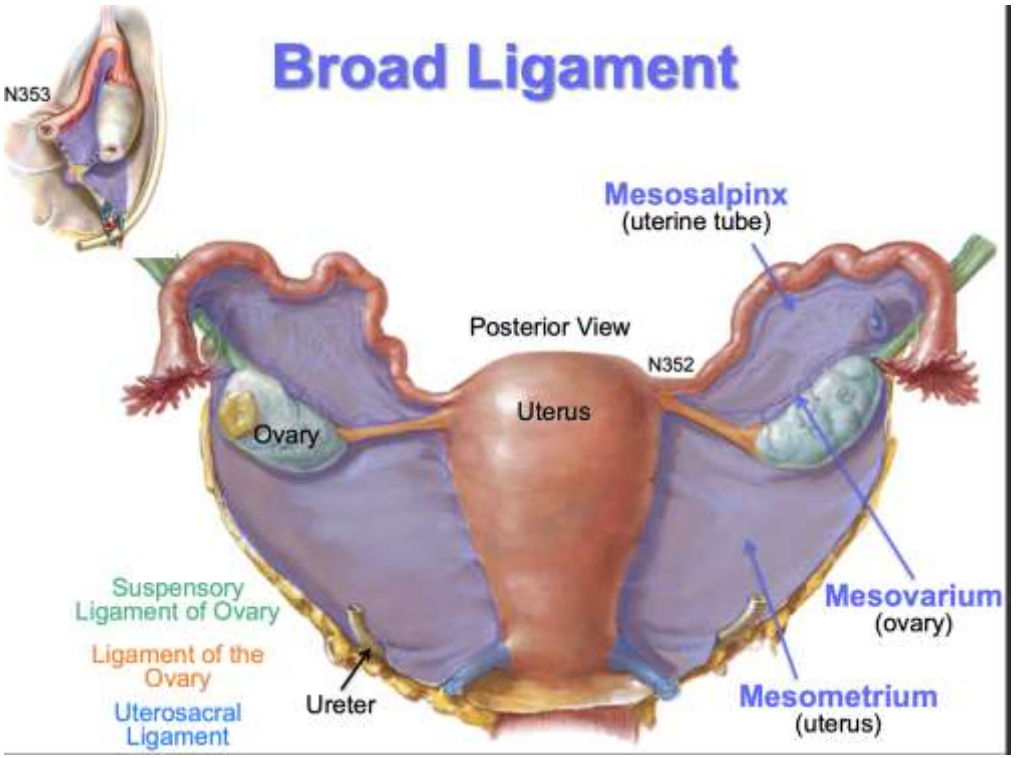




Uterine Dynamic Support of the uterus is provided by the **pelvic diaphragm** and **pelvic floor** musculature. Its tone during sitting and standing and active contraction during periods of increased intraabdominal pressure (coughing, sneezing, etc.) is transmitted through the **endopelvic fascia** in which they are embedded.

Uterine Passive Support of the uterus is provided by its anteverted and anteflexed position over the bladder. An increase in intra abdominal pressure presses the uterus against the bladder. It also has many ligamentous supports.

a. The cervix is the least mobile portion of the uterus because condensations of **endopelvic fascia** attach to the cervix providing passive support:
   i. **Cardinal (transverse cervical) ligament** – located at the base of the broad ligament, extending from the cervix and vagina to the lateral pelvic walls. Contains the **uterine vessels** and helps hold the uterus in position
   ii. **Uterosacral ligaments** – pass from the cervix to the mid-sacrum; provide support for the uterus to keep it from prolapsing into the vagina



Other Uterine Ligaments:

a. **Broad ligament of the uterus** – double layer of peritoneum extending from the sides of the uterus to the lateral pelvic walls and floor. Formed from the most inferior aspect of the peritoneum draping over the pelvic organs. This ligament assists with keeping the uterus in position. Consists of three parts:
   i. **Mesometrium** – largest part (inferior to mesosalpinx and mesovarium); covers uterus
   ii. **Mesosalpinx** – covers uterine tubes
   iii. **Mesovarium** – covers ovaries

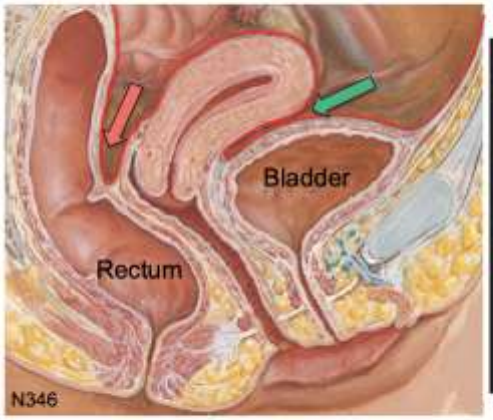
b. **Round ligament of the uterus** – attaches anteroinferiorly to the uterotubal junction of the uterus and runs within the broad ligament. Exits the pelvis via the deep inguinal ring and inserts into the labium majora (homologous to the gubernaculum)

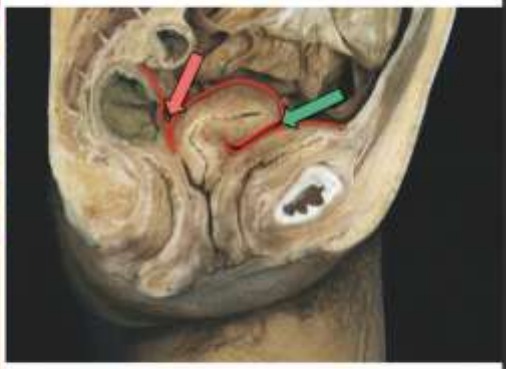
Ovarian Support:

a. **Suspensory ligament of the ovary** – attaches the ovary to the lateral pelvic wall and carries the ovarian vessels; essentially it is where the peritoneum is draped over the ovarian vessels and thus becomes continuous with the mesovarium medially

b. **Ligament of the ovary** – attaches the ovary to the lateral edge of the uterus, inferior to the uterine tube (was the cranial portion of the gubernaculum)

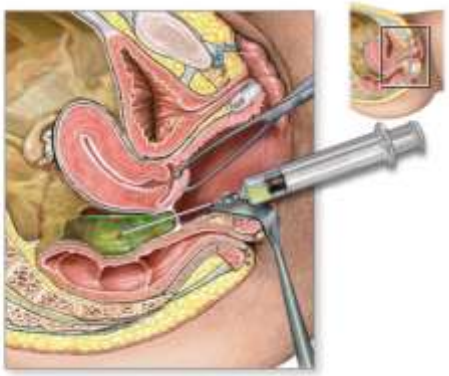
Recall from P1: Peritoneal Recesses





Access to Pelvic & Peritoneal Cavity

- Minimal tissue between peritoneal cavity (rectouterine pouch) or the pelvic cavity and the external environment (vaginal canal)
- May be an access point for:
  - Culodcentesis
  - Culdoscopy



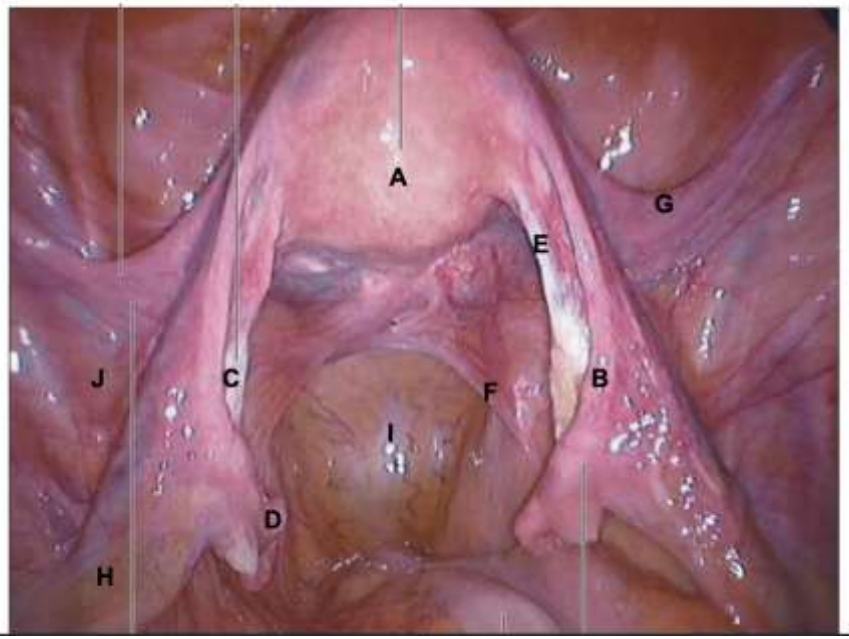
Peritoneal Recesses:

a. **Vesicouterine pouch** – where the peritoneum is reflected from the uterus onto the posterior margin of the superior surface of the bladder
b. **Rectouterine pouch** – where the peritoneum is reflected from the uterus onto the anterior aspect of the rectum. This is the inferior-most extent of the pelvic cavity in females in an upright position. Surgically accessible via the posterior fornix.

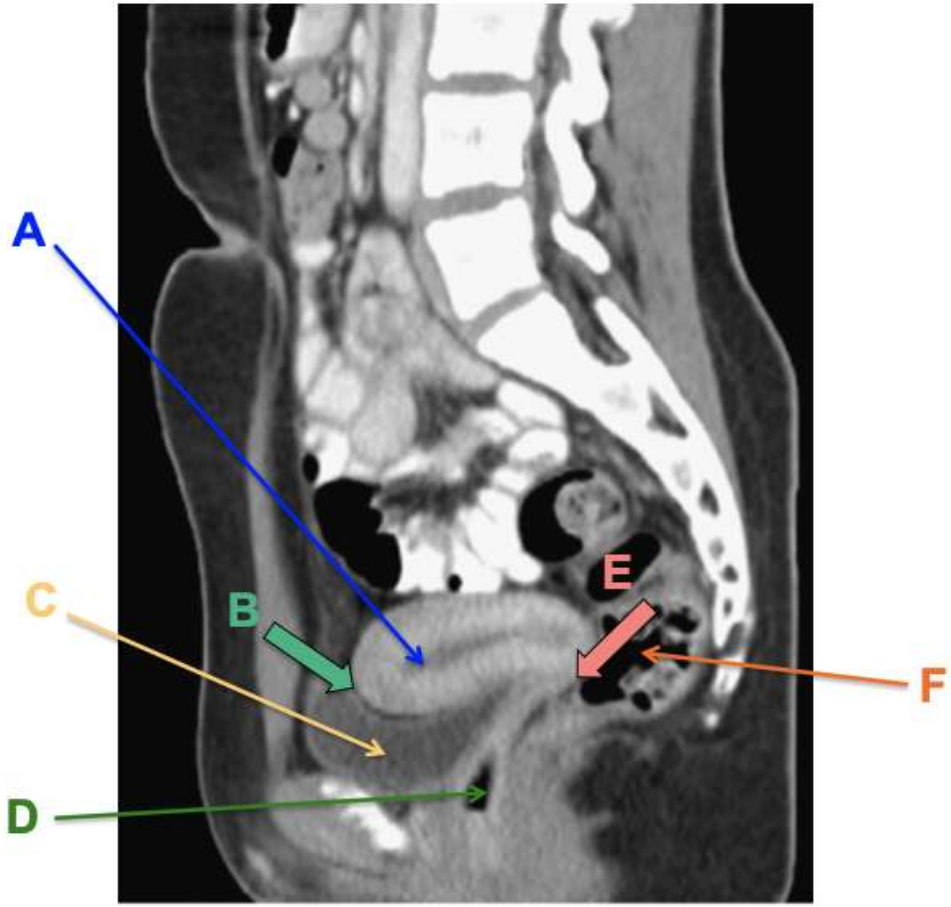
Clinical Note: The posterior fornix is the deepest portion and directly related to the rectouterine pouch. This constitutes the area where there is minimal tissue intervening between the peritoneal cavity and the external environment. It is used as a site for culdocentesis and culdoscopy.

Knowledge Check

Match the correct structure name to the label in the female pelvis.



Review



Uterus	A
Fallopian Tube	B
Ovary	C
Fimbrae	D
Ligament of the Ovary	E
Uterosacral Fold (Ligament)	F
Round Ligament of the Uterus	G
Suspensory Ligament of the Ovary	H
Rectouterine Pouch	I
Broad Ligament	J

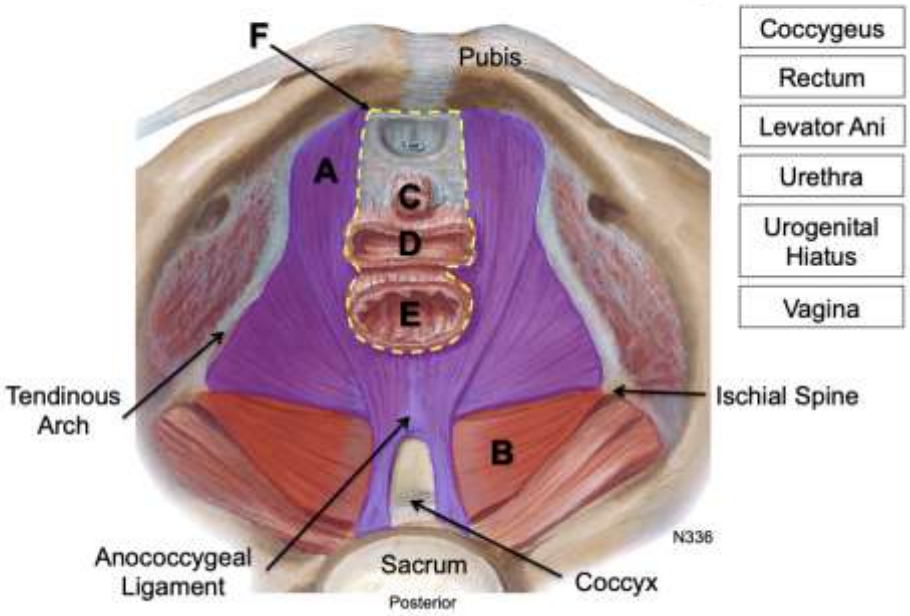
Uterus	A
Vesicouterine Pouch	B
Bladder	C
Vagina	D
Rectouterine Pouch	E
Rectum	F

The muscles, ligaments, and fascia of the pelvis provide dynamic and passive support to the organs of the pelvis. If these supports fail, women may experience complications such as continence or prolapse issues.

PELVIC FLOOR

Recall from P1: Pelvic Floor

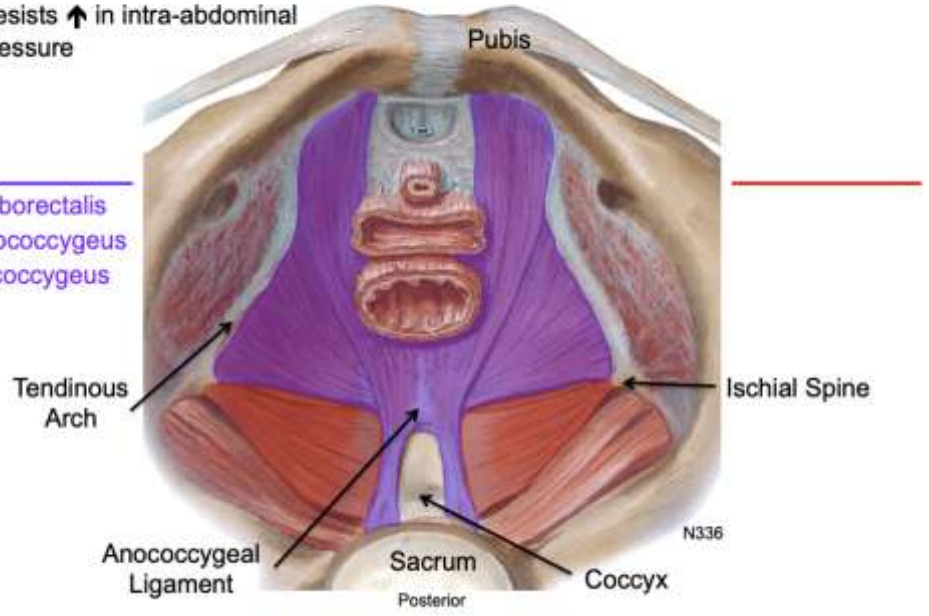
Match the correct structure to each label on the pelvic floor



Coccygeus
Rectum
Levator Ani
Urethra
Urogenital Hiatus
Vagina

Recall from P1: Pelvic Floor

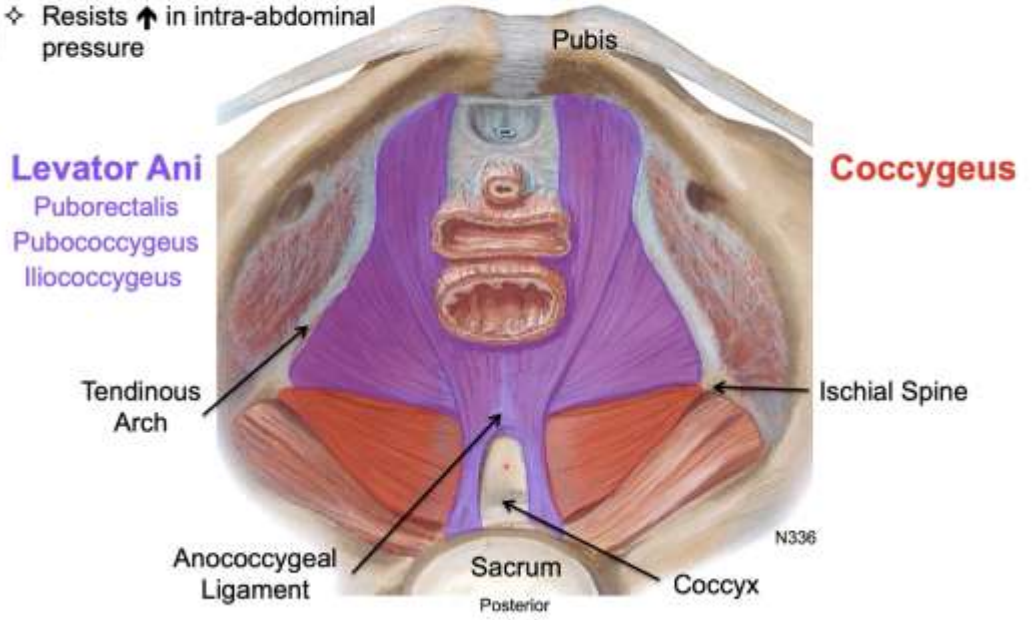
- Supports abdominopelvic viscera
- Resists ↑ in intra-abdominal pressure

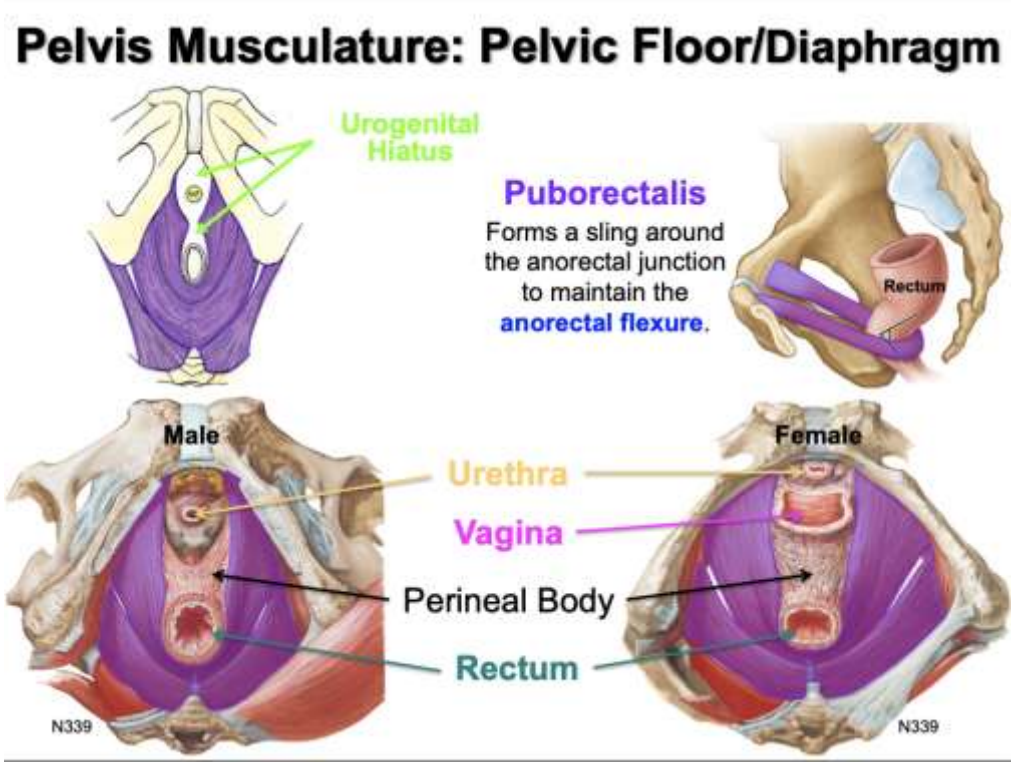


- Levator Ani → A
- Coccygeus → B
- Urethra → C
- Vagina → D
- Rectum → E
- Urogenital Hiatus → F

Recall from P1 #2: Pelvic Floor

- Supports abdominopelvic viscera
- Resists ↑ in intra-abdominal pressure





Pelvic Floor:

1. The pelvic diaphragm (floor) supports the pelvic viscera and consists of two muscles:
    a. **Coccygeus** – travels from the ischial spine to the inferior sacrum and coccyx
    b. **Levator Ani** – travels from the pubic body, tendinous arch of o fascia, and ischial spine to the perineal body, coccyx, and anococcygeal ligament; it consists of three parts
        i. Puborectalis medially
        ii. Pubococcygeus intermedially
        iii. Iliococcygeus posterolaterally

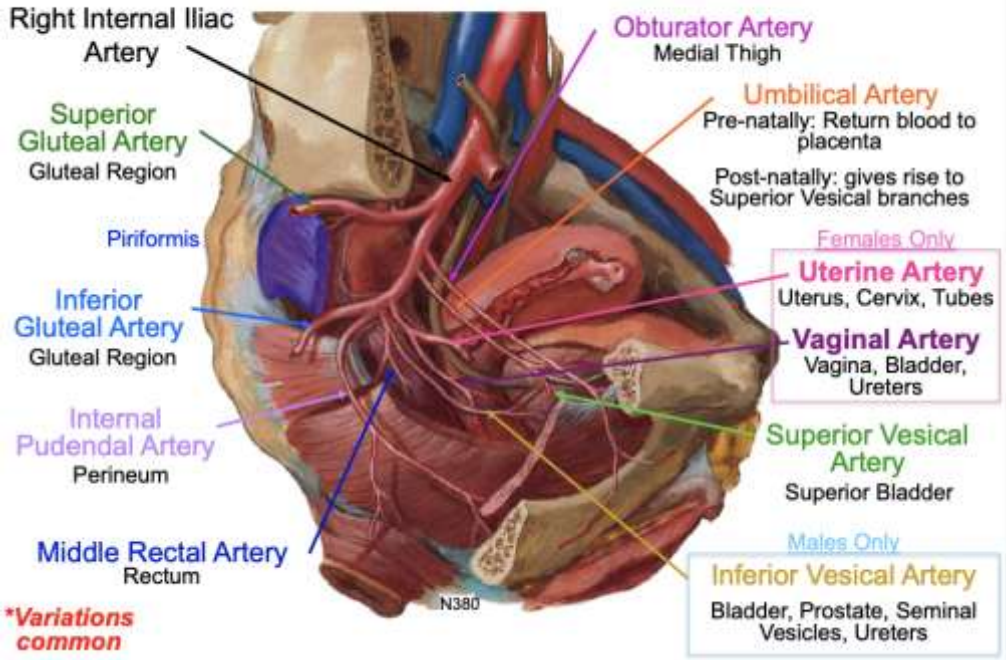
How did you do?

If you have difficulty recalling this content, please go back and review the content from P1.

- The pelvic floor is the inferior most aspect of the pelvis and the roof to the perineum
- The pelvic floor is integral in maintaining intraabdominal pressure and supporting the pelvic viscera

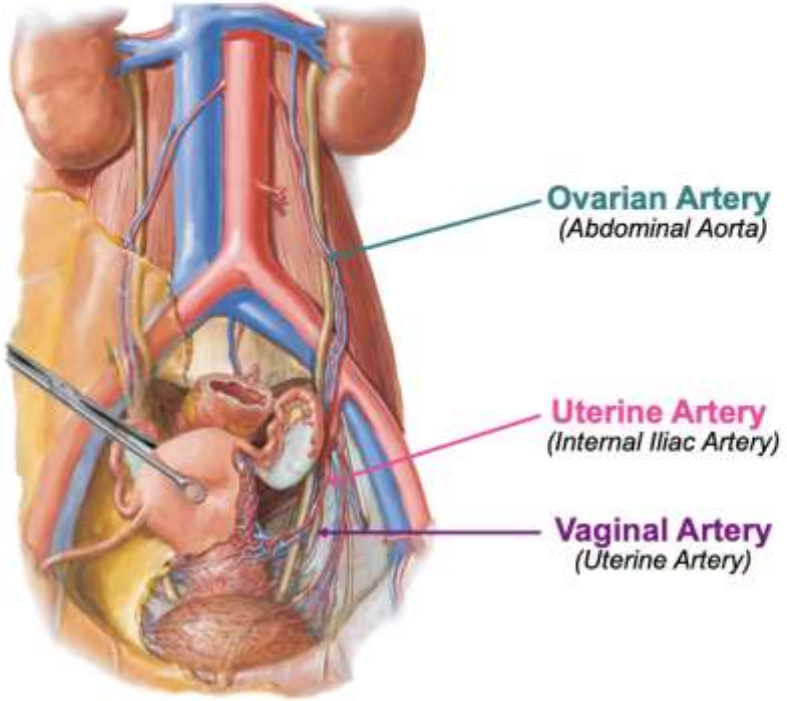
VASCULAR SUPPLY & LYMPHATIC DRAINAGE

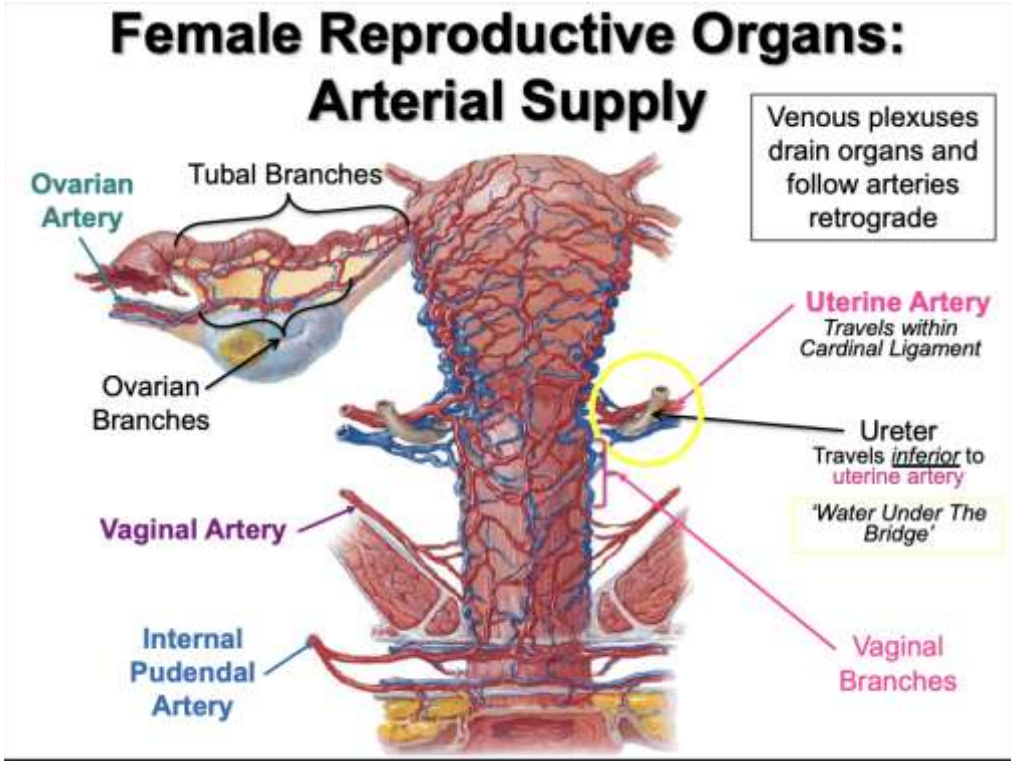
Review: Pelvic Arteries



Variations common

Female Reproductive Organs: Arterial Supply





Arteries:

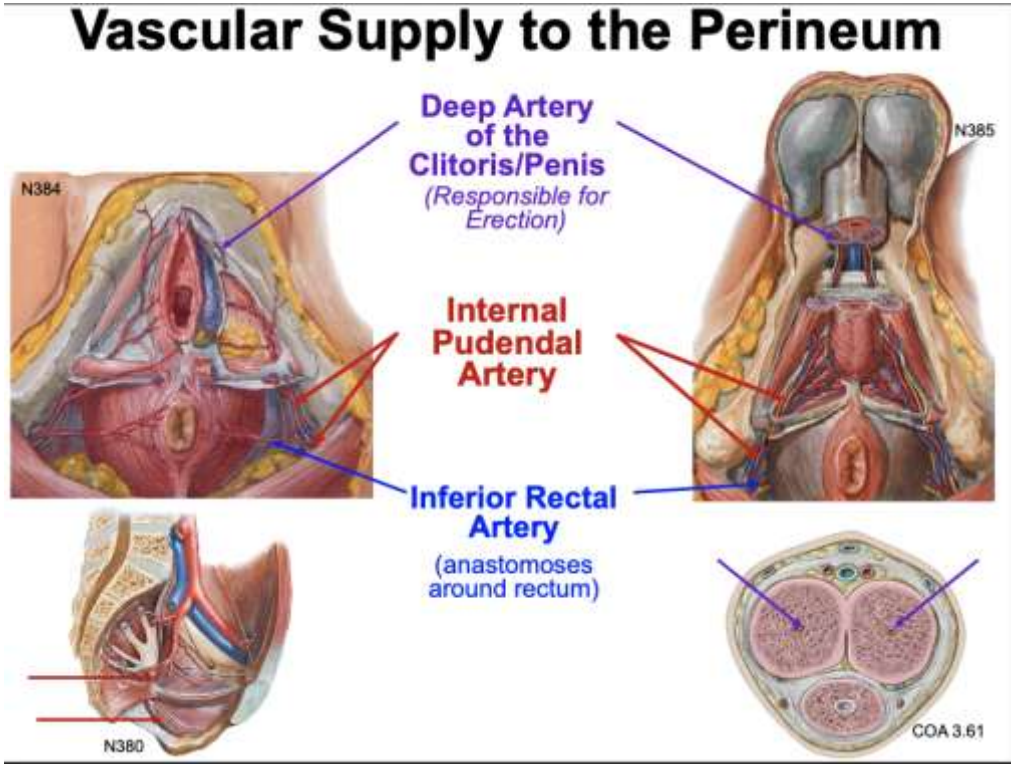
a. Review the branches of the internal iliac artery from GU
b. **Uterine arteries** – branch of internal iliac arteries; runs medially toward the cervix; pass superior to the ureters near the lateral fornix of the vagina; travel within the cardinal ligaments

Clinical Note: The relationship between the ureters and the uterine artery is important (water under the bridge) for surgeons ligating the uterine artery during a hysterectomy. The ureters must be positively identified and excluded from ligature.

a. **Vaginal branches** descend to supply the cervix and superior vagina
b. Ascending branches course along the lateral side of uterus to reach the medial aspects of the ovaries and uterine tube, terminating into ovarian (to ovary) and tubal (to uterine tube) branches
c. **Vaginal artery** (homolog of male inferior vesical artery) supplies the middle aspect of the vagina

Ovarian arteries – branch of the abdominal aorta; pass through **suspensory ligament** of the ovary to reach lateral aspects of the ovaries and the uterine tubes

a. Divide into **tubal** and **ovarian branches**
    i. **Tubal** and **ovarian** branches of the **uterine** and **ovarian arteries** anastomose with each other (collateral circulation)

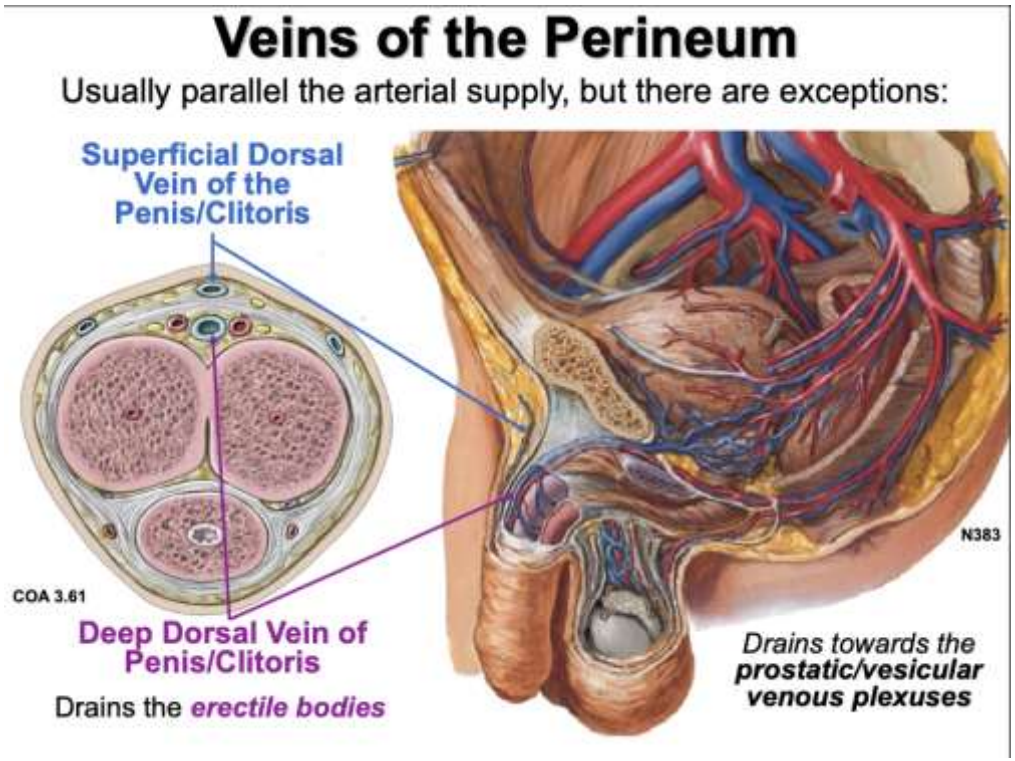


Internal pudendal artery – supplies the inferior vagina

a. Exits pelvis via the **greater sciatic foramen**, passes post. to **sacrospinous ligament**, then enters perineum through **lesser sciatic foramen**
b. Gives off many branches to the perineal region. Some branches to note:
    i. **Inferior Rectal (Anal) Artery** - Branches within the pudendal canal; supplies the anal canal & perianal skin.
c. **Deep (central) Artery of the Penis (Clitoris)** - A terminal branch of internal pudendal a. Enters the penile/clitoral crura of corpora cavernosa. Supplies most of the blood to the corpora cavernosa and is primarily responsible for erection. Most ***helicine (spiral) arteries*** are terminal branches of this artery.

Clinical Note: Due to the prevalence of uterine and cervical cancer, **hysterectomy** (excision of the uterus) is a relatively common procedure. The uterus may be approached and removed surgically through the anterior abdominal wall or through the vagina. During the procedure, tremendous care is taken to identify the distal ureters and ligate the nearby uterine arteries without damage to the ureters.

Clinical Note: The collateral circulation around the female reproductive organs allows the ovaries to be left in place when the uterine artery is ligated during a hysterectomy, to avoid the patient requiring estrogen therapy.

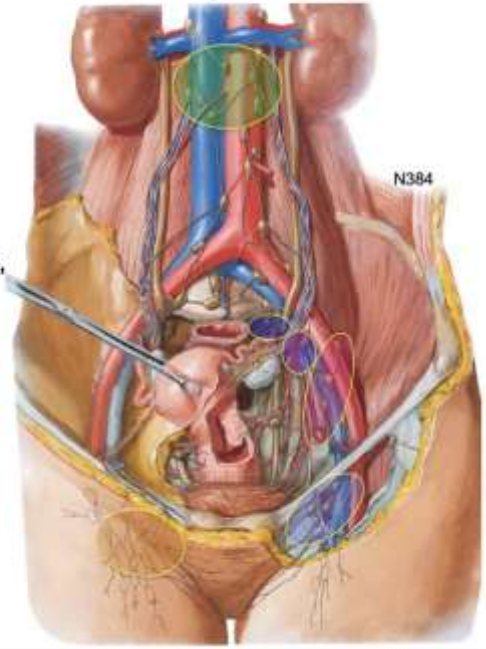


Venous Drainage: A plexus of veins drain the reproductive organs. Veins then follow the arteries retrograde to the internal iliac veins and inferior vena cava.

a. **Veins of the Perineum** – usually parallel the arterial supply, but there are exceptions:
   - **i. Superficial Dorsal Vein of the Penis**
     1. Drains skin & subcutaneous tissue of the penis.
     2. Usually unpaired (but variable)
     3. Empties primarily into the greater saphenous vein via the external pudendal veins.
     4. Can anastomose with the deep dorsal vein; thus blood from the skin & subcutaneous structures can reach the prostatic venous plexus.
   - **ii. Deep Dorsal Vein of Penis (clitoris)**
     1. Drains most of the blood from the glans penis, prepuce, & erectile tissues.
     2. ***Drains the erectile bodies*** through a gap between the inferior pubic symphysis (arcuate pubic ligament) & perineal membrane into the prostatic ***venous plexus*** (males) or ***vesicular plexus*** (females).

Lymphatic Drainage

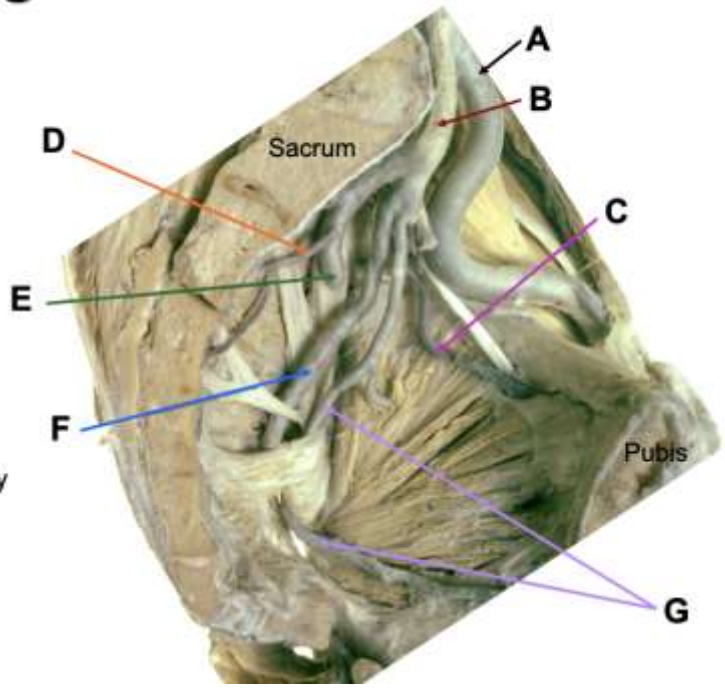
- **Superficial Inguinal Lymph Nodes:**
  - Drainage from skin
- **Deep Inguinal Lymph Nodes:**
  - Drainage from erectile bodies and external genitalia
- **External Iliac Lymph Nodes:**
  - drainage from upper vagina, cervix, & inferior uterine body
- **Internal Iliac Lymph Nodes:**
  - drainage from mid-upper vagina, cervix, uterine body, & male cavernous bodies
- **Sacral Lymph Nodes:**
  - drainage from the inferior vagina
- **Lumbar/Pre-aortic Lymph Nodes:**
  - drainage from the ovaries, uterine tubes, & uterine fundus



Lymphatic Drainage: For the most part the lymphatic drainage reflects the venous pathway (with some exceptions):

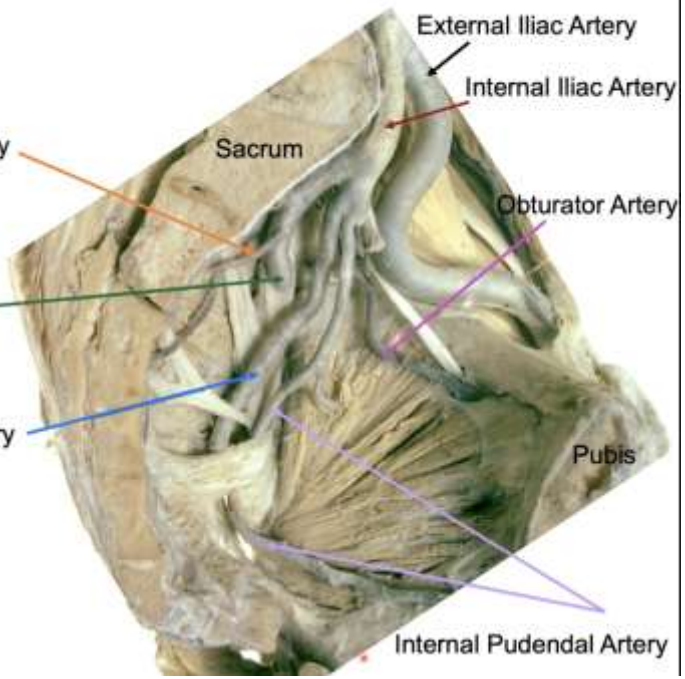
a. ***Lumbar (Pre-arotic) Lymph Nodes*** drain the ovary, uterine tubes, and the fundus of the uterus
b. ***Internal Iliac Lymph Nodes*** drain the middle and upper vagina, cervix and body of the uterus, and male cavernous bodies (males)
c. ***External Iliac Lymph Nodes*** drain the upper vagina, cervix and lower body of uterus
d. ***Sacral Lymph Nodes*** drain the inferior vagina
e. ***Superficial inguinal lymph nodes*** – drainage from skin (scrotum, labia, penis, & prepuce)
f. ***Deep inguinal lymph nodes*** – drainage from the clitoris, vestibular bulb, anterior labia minora, glans penis, & penile urethra

Knowledge Check: Pelvic Arteries



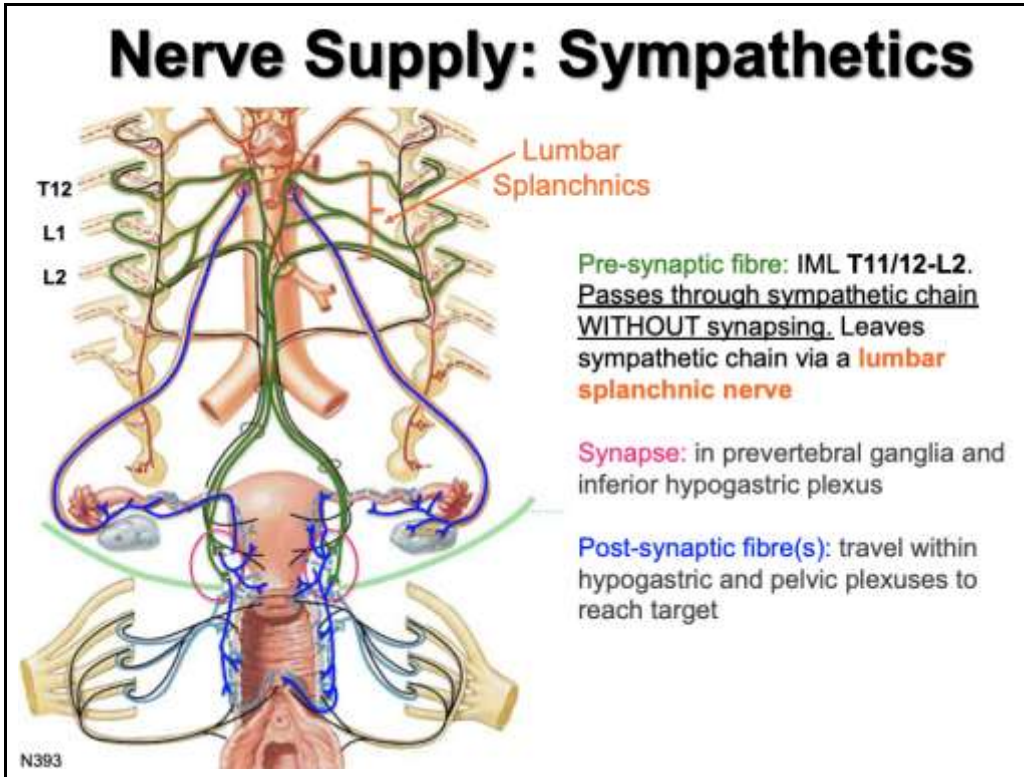
Inferior Gluteal Artery
Internal Iliac Artery
Internal Pudendal Artery
External Iliac Artery
Lateral Sacral Artery
Obturator Artery
Superior Gluteal Artery

Knowledge Check: Pelvic Arteries



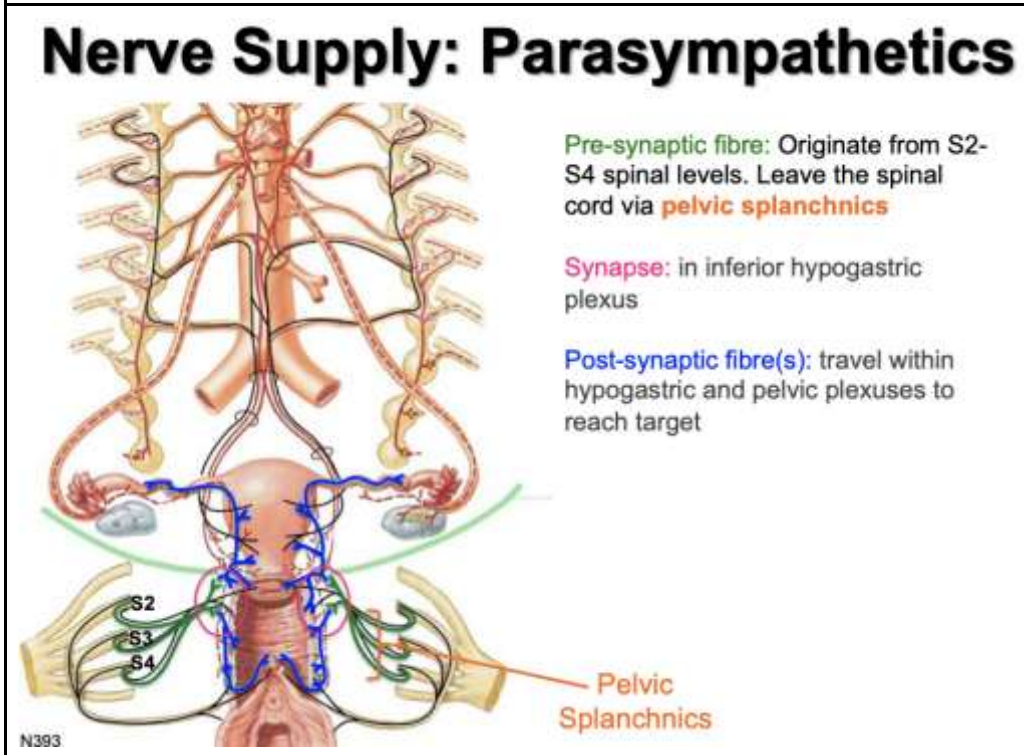
- Blood supply to the pelvis and perineum are supplied by branches of the internal iliac artery. These are arteries are named for where they are going and not where they are branching from.
- The uterine artery is the primary blood supply to the uterus. It is also in close relationship with the ureter in the pelvis as the ureter courses anteriorly to the bladder.
- Skin, erectile tissues, reproductive organs, and the gonads all have different patterns of lymphatic drainage.
- The vascular supply and lymphatic drainage of the ovaries is representative of its developmental origin in the posterior abdominal wall.

NERVE SUPPLY



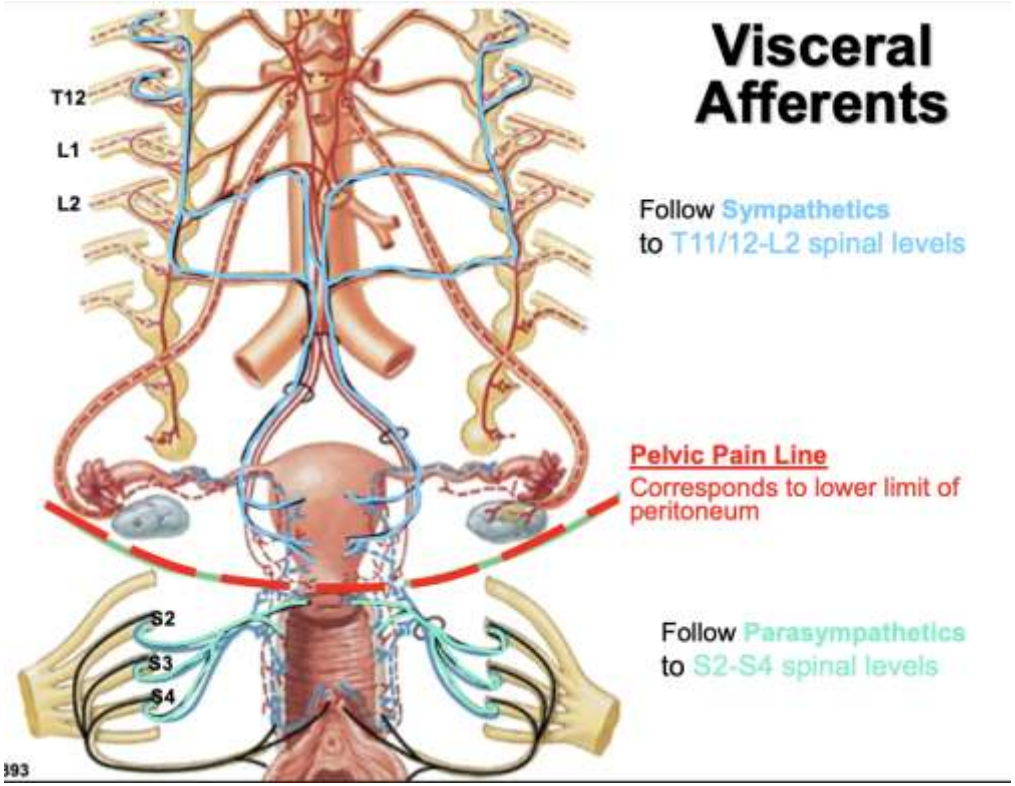
Sympathetic Innervation:

a. ***Presynaptic sympathetic fibres*** originate in the intermediolateral (IML) horns of spinal levels ***T11/12-L2*** and they will pass through the sympathetic chain without synapsing. They leave the sympathetic chain via lumbar splanchnic nerves.
b. ***Synapse:*** Presynaptic fibres synapse in prevertebral ganglia and the hypogastric plexuses with postsynaptic fibres
c. ***Postsynaptic sympathetic fibres*** travel through the superior and inferior hypogastric plexuses to reach the female reproductive pelvic viscera
    i. Sympathetic activity may produce uterine contraction and vasoconstriction, but these activities are complicated by hormonal control of reproductive organ functions



Parasympathetic Innervation:

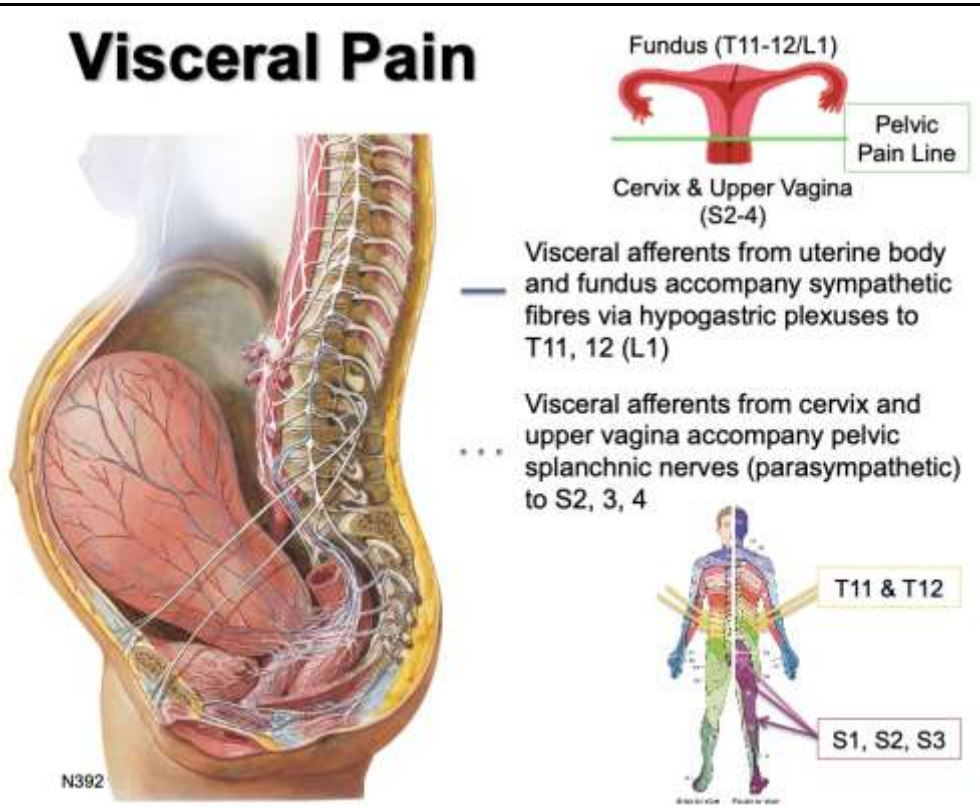
a. ***Presynaptic sympathetic fibres*** originate in the ***S2-S4*** spinal segments and pass to the inferior hypogastric and pelvic plexuses via pelvic splanchnic nerves
b. ***Synapse*** in ganglia, within the inferior hypogastric and pelvic plexuses
c. ***Postsynaptic sympathetic fibres*** are comparatively short neurons that travel from the pelvic plexuses to the female reproductive viscera
    i. Parasympathetic activity may produce uterine inhibition and vasodilation, but these activities are complicated by hormonal control of reproductive organ functions



Above the pelvic pain line – The fundus and body of the uterus, the uterine tubes, and the ovaries are above the pelvic pain line; thus pain sensation travels retrogradely with the sympathetic pathways to **T11/12-L2 dorsal root ganglia (DRG).** The brain interprets pain from these organs as coming from the T11/12-L2 dermatomes (mid-trunk, back, and flank).

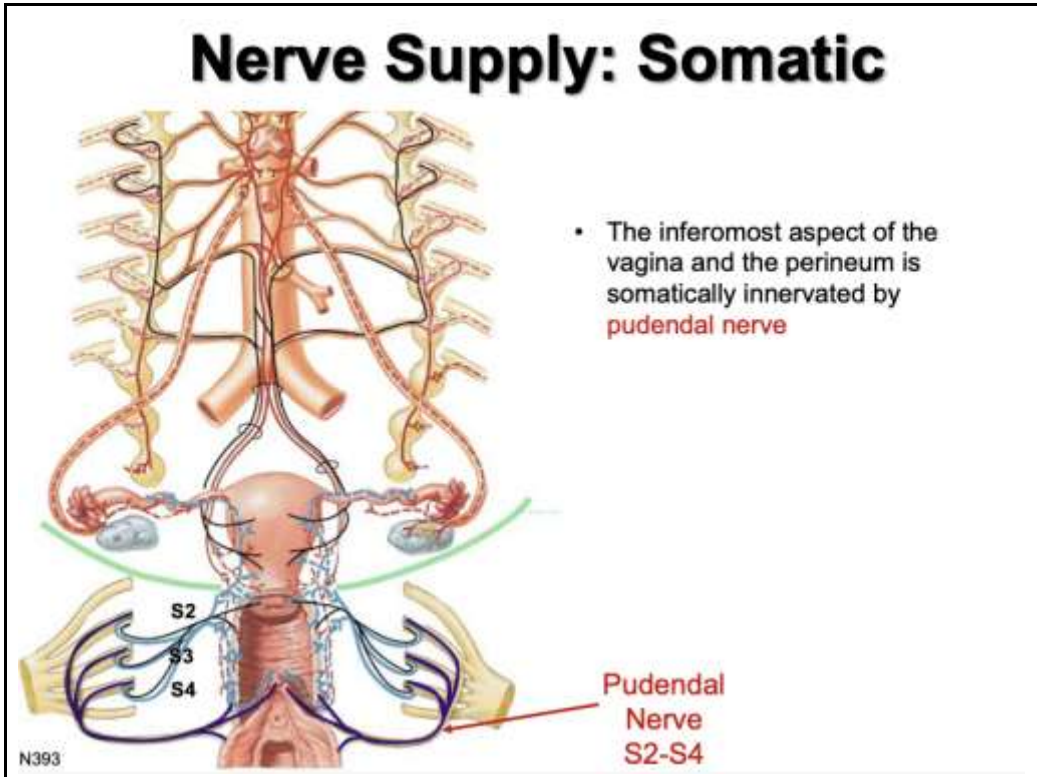
Below the pelvic pain line – The vagina and uterine cervix are below the pelvic pain line; thus pain sensation travels retrogradely with the **parasympathetic** pathways to the **S2-S4 DRG**. The brain interprets pain from these organs as coming from the S2-S4 dermatomes (posterior aspect of the gluteal region, thigh, and leg).

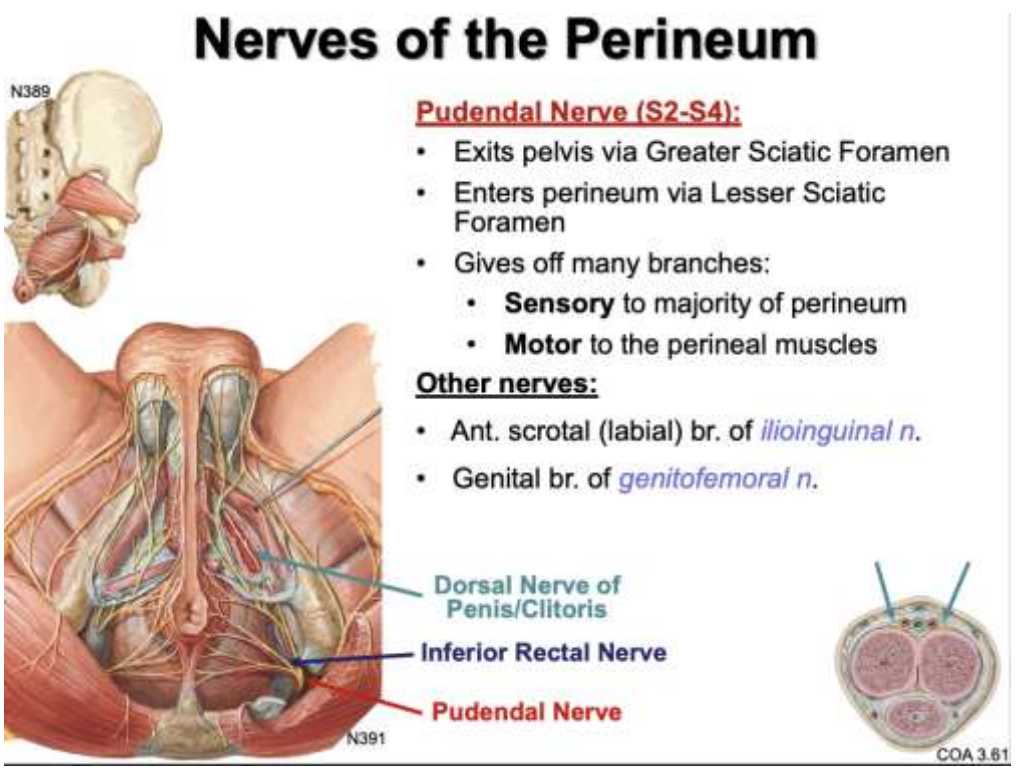
a. **Clinical Note:** Uterine contractions (during menses and childbirth) are hormonally stimulated; not the result of nerve stimulation



Visceral Pain

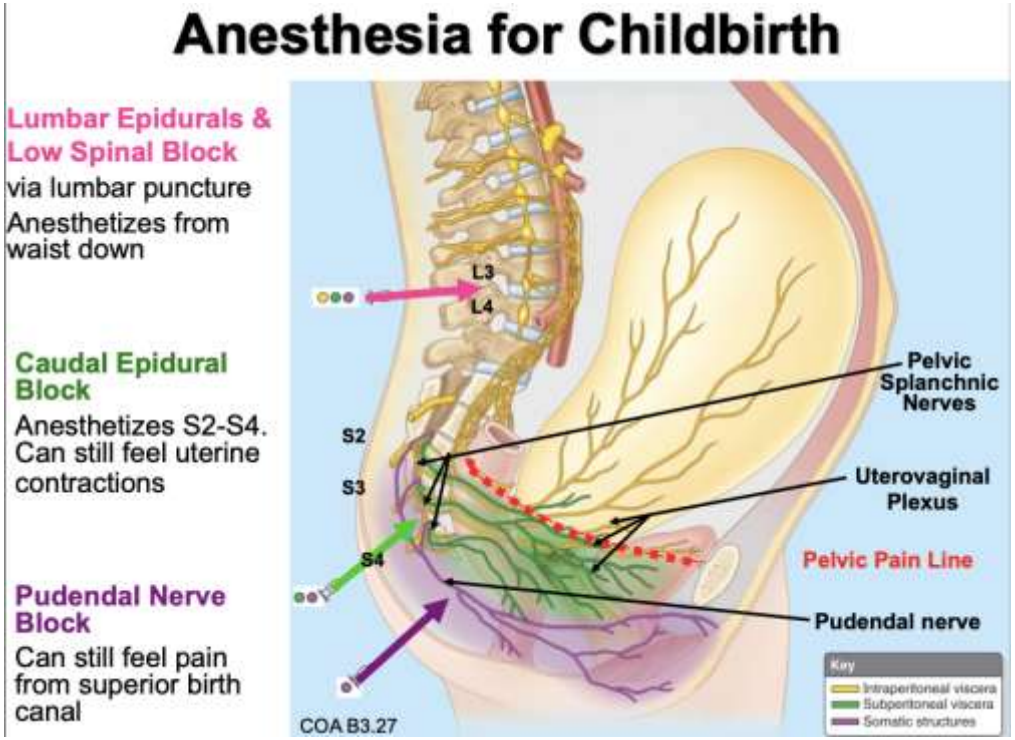
a. Fundus is above the pain line
    i. Visceral afferents from body and fundus (referred to T11,12, (L1)
b. Cervix and Upper vagina is below the pain line
    i. Visceral afferents from cervix and upper vagina (referred to S2,3,4)





Pudendal Nerve (S2-S4): Exits pelvis via the **greater sciatic foramen**, passes post. to **sacrospinous ligament**, then enters perineum through lesser **sciatic foramen**, then branches as it enters the pudendal canal.

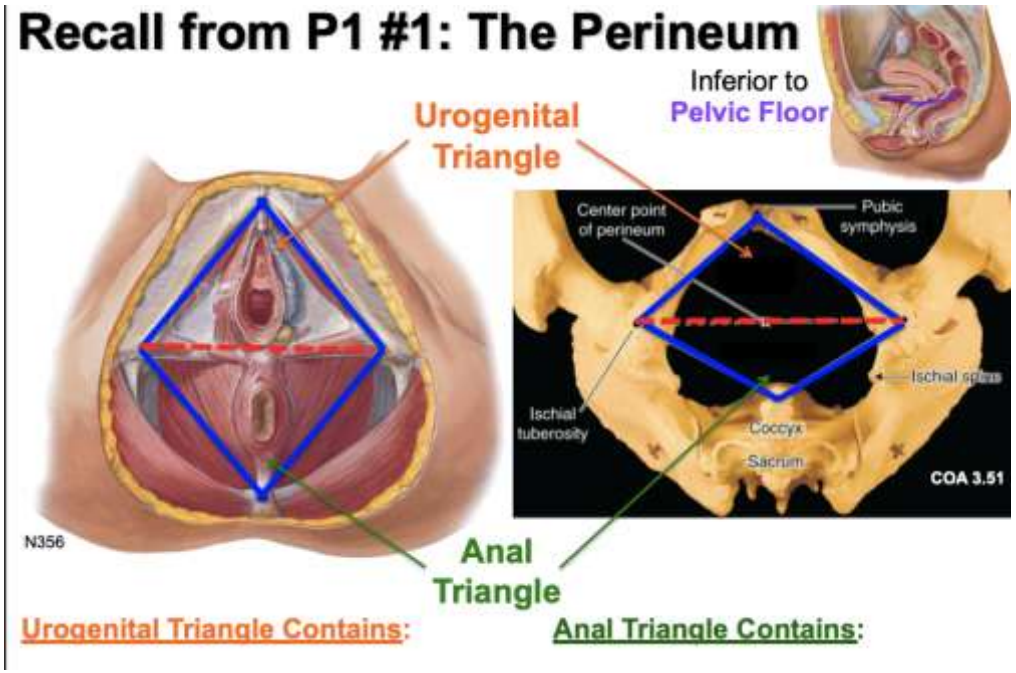
a. Provides Motor to perineal muscles & sensory to majority of perineal region via its branches. Some important branches to note:
b. **Inferior Rectal (Anal) Nerve (S3-S4)** - Arises at entrance to pudendal canal to traverse ischioanal fat to reach anal canal to supply the anal canal (inf. to pectinate line): external anal sphincter, mucosa, & skin
c. **Dorsal Nerve of the Penis/Clitoris** - Terminal branch of pudendal nerve, enters the deep perineal pouch where it supplies the muscles in this space. Passes to dorsum of penis, runs lateral to dorsal arteries supplying skin & glans. Primary sensory nerve to the penis & clitoris



Clinical Note: Anesthesia of the pelvis and perineum is often used in childbirth

a. **Lumbar epidurals and low spinal blocks** – anesthesia is administered at the L3/4 level, which completely anesthetizes the lower half of the body (waist down)
b. **Caudal (saddle) epidural nerve block** – anesthesia is administered via an indwelling catheter in the sacral canal. It anesthetizes the lower sacral segments (S2-S4) and thus structures innervated by the pudendal and pelvic splanchnic nerves, but not the uterine fundus. It is often administered to patients electing participatory childbirth techniques (Lamaze, etc.), since they can still feel uterine contractions and know when to bear down
c. **Pudendal nerve block** – anesthesia is administered into the tissues surrounding the pudendal nerve through the skin or via the vagina at the ischial spine; it anesthetizes most of the perineum and inferior quarter of the vagina. However, it does not black pain from the superior birth canal (uterine cervix and superior vagina), so the mother can feel uterine contractions.

REVIEW of PERINEUM from P1



Review of Perineum from P1:

a. Diamond-shaped compartment inferior to the pelvic diaphragm (floor)
   i. The overlying surface area is the perineal region.

b. General Boundaries:

   i. **Bony Boundaries** (pelvic outlet):
      1. Tip of the coccyx
      2. Ischial tuberosities (and sacrotuberous ligaments)
      3. Ischiopubic rami
      4. Pubic Symphysis

   ii. **Superficial boundaries** (in the lithotomy position):
      1. ***Anteriorly:*** Females: mons pubis; Males: pubic region
      2. ***Laterally:*** medial surface of thighs
      3. ***Posteriorly:*** gluteal folds & superior end of intergluteal cleft

c. A transverse line between the ischial tuberosities divides the perineum into:

   i. **Urogenital (UG) Triangle** (anterior triangle – oriented almost horizontally)
      1. contains the external genitalia and related structures

   ii. **Anal Triangle** (posterior triangle – oriented on an angle)
      1. contains the anal canal and associated musculature

Recall from P1 #2: Fascial Planes





17 year old male presents to the ED after a straddle injury with swelling in his groin region.

What important structure may be damaged?

_____

Where would urine and blood accumulate?

_____

Recall from P1 #3: Fascial Planes

26 year old male presents to the ED after a car accident. A FAST scan indicates free fluid in the inferior peritoneal cavity.



What important structure may be damaged?

_____

Where would urine and blood accumulate?

_____

Why would it be in the peritoneal Cavity?

_____

_____



Recall from P1 #2: Fascial Planes





17 year old male presents to the ED after a straddle injury with swelling in his groin region.

What important structure may be damaged?

Spongy (penile) Urethra

Where would urine and blood accumulate?

Superficial Perineal Pouch

Recall from P1 #3: Fascial Planes

26 year old male presents to the ED after a car accident. A FAST scan indicates free fluid in the pelvic cavity.



What important structure may be damaged?

Membranous Urethra

Where would urine and blood accumulate?

Deep Perineal Pouch

Why would it be in the pelvic cavity?

Fluid can pass superiorly
through the UG Hiatus



Review of Perineum from P1:

a. **Perineal Membrane** – thin sheet of deep fascia covering the urogenital diaphragm
   i. Perforated by the urethra in both sexes as well as the vagina in females
   ii. Provides attachment for erectile bodies and associated muscles

b. Perineal Body

i. Central point of the perineum, midway between the ischial tuberosities

ii. Musculofibrous mass of connective tissue lying deep to the skin

iii. Developmental origin from fusion site of urorectal septum with cloacal membrane.

1. **Clinical Note:** The perineal body is important clinically in the female because it is the final support of the pelvic viscera, linking muscles that extend across the pelvic outlet. The attachment points of perineal muscles can be stretched or torn during childbirth, thereby losing support of the pelvic floor. This can lead to prolapse of pelvic viscera.
2. **Clinical Note: Episiotomy:** surgical incision of the perineum & vaginal wall. Performed to enlarge the vaginal orifice, with the intention of decreasing excessive traumatic tearing of the perineum.
   a. During a ***median episiotomy,*** the perineal body is incised. While this is thought to limit difficulties from scarring and limit further tearing, when tearing does happen, it is directed toward the anus. Which results in an increased incidence of severe lacerations, long-term incontinence, pelvic prolapse, & anovaginal fistulae.
   b. ***Mediolateral episiotomies*** appear to result in a lower incidence of severe lacerations & are less likely to be associated with damage to the anal sphincters & canal.

c. Perineal Fasciae:

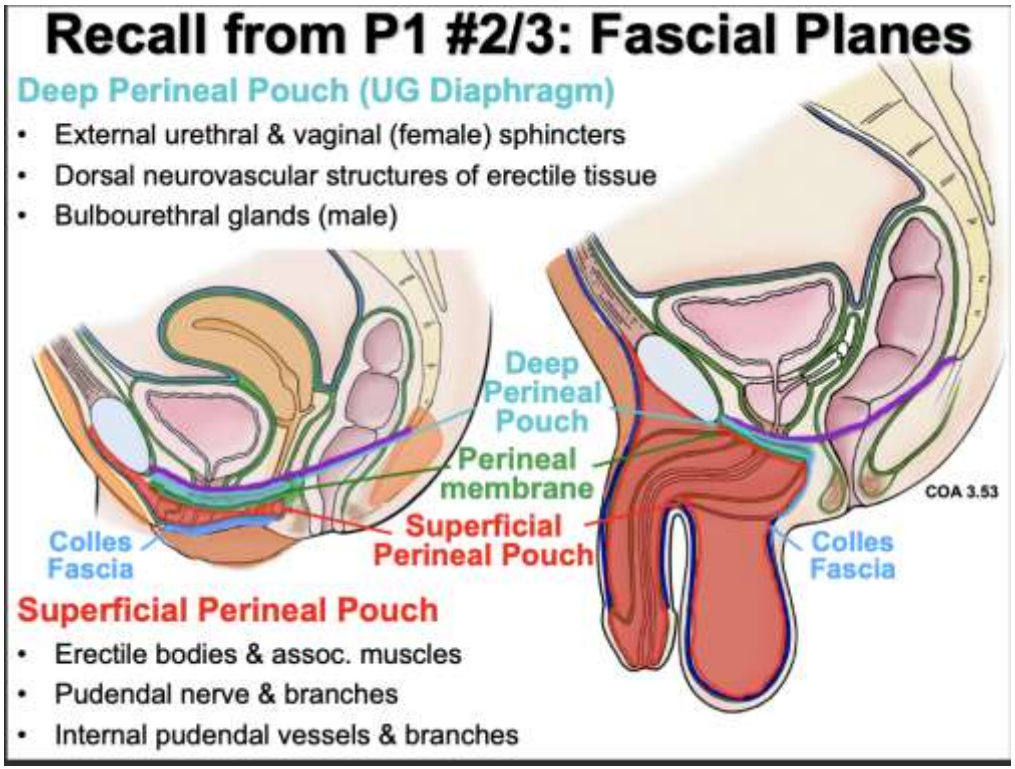
i. Superficial Perineal Fascia:

1. Superficial Fatty layer
   a. Continuous with the fatty layer of subcutaneous tissue of the abdomen (Camper's Fascia)
   b. **Males:** diminished in males, replaced by dartos smooth muscle in the penis and scrotum
   c. **Females:** makes up the substance of the labia majora and mons pubis
   d. Continuous posteriorly with the ischioanal fat pad
2. Deep Membranous Layer (**Colles Fascia**)
   a. Downward continuation of the membranous layer of superficial fascia of the abdomen (***Scarpa's fascia***) in both males & females
   b. Limited to the UG triangle, does not extend into the **anal** triangle
   c. Attached laterally to the ***fascia lata*** of the thigh
      i. **Males:** Continuous anteriorly with the dartos fascia of the penis & scrotum
         1. On each side of & anterior to the scrotum, continuous with Scarpa's fascia in abdomen
      ii. **Females:** Passes superior to the fatty layer in the labia majora
      iii. Superiorly continuous with Scarpa's fascia of the abdomen

ii. Deep Perineal Fascia: Investing/Gallaudet Fascia

1. Invests the **ischiocavernosus, bulbospongiosus, & superficial transverse perineal muscles**

2. Attached laterally to ischiopubic rami
3. Fused anteriorly to the suspensory ligament of the penis or clitoris
4. Continuous with the deep fascia of the external oblique muscle & rectus sheath



Perineal Pouches:

a. **Superficial Perineal Pouch:**
   i. ***Potential space*** between the membranous layer of subcutaneous tissue of the perineum (Colles fascia) & ***perineal membrane***; bounded laterally by the ischiopubic rami.
   ii. Contains:
      1. Erectile bodies and associated muscles
      2. Pudendal nerve, internal pudendal vessels and their branches
   iii. **Clinical Note:** When the male urethra is ruptured possibly due to a ***straddle injury*** or incorrect insertion of a catheter, there is accumulation of blood and or urine (**urinary extravasation**) into the superficial perineal pouch. Fluid cannot extend to the thigh or anal triangle so it moves upward onto the anterior abdominal wall underneath the skin and Camper's & Scarpa's fascia, but external to the external abdominal oblique muscle layer. (***COA p. 415-6***)

b. **Deep Perineal Pouch:**
   i. Located between the **perineal membrane** (floor) & inferior fascia of the pelvic diaphragm (roof) and is bounded laterally by the inferior portion of the fascia of the obturator internus.
      1. Between these two membranes lies an envelope of deep fascia and surrounding muscles commonly referred to as the

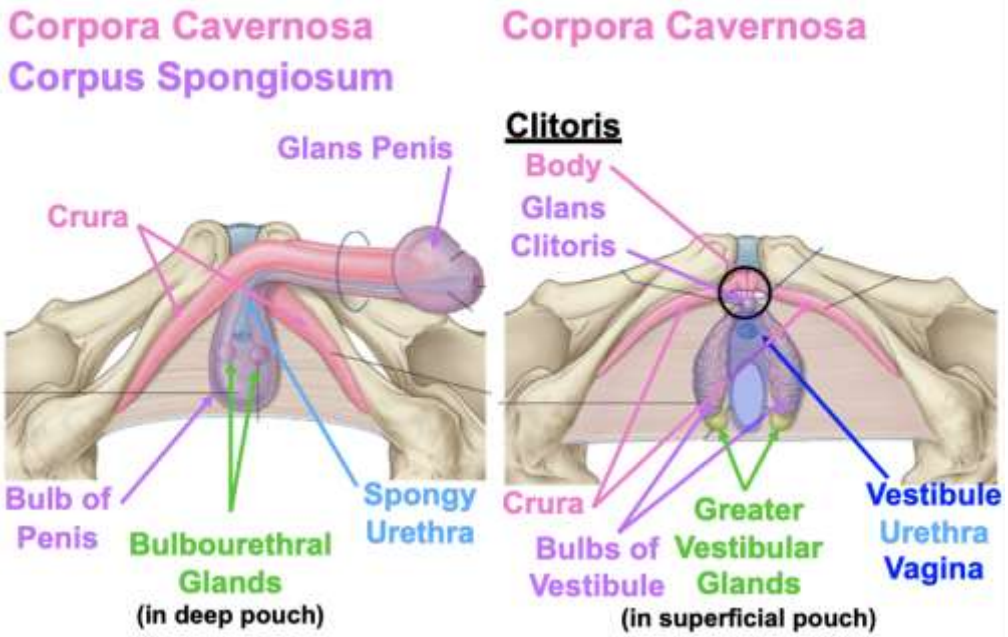
urogenital diaphragm

a. ***Males:*** the main part of the muscle forms the **external urethral sphincter**, which surrounds the membranous urethra as it passes from the prostate gland into the perineum and is under voluntary control. Contains bulbourethral glands, embedded within the deep perineal musculature and dorsal neurovascular structures of the penis
b. ***Females:*** both the vagina and urethra pass through the muscle, which acts as a sphincter for both. The **external urethral sphincter** is voluntary (somatic innervation). Contains dorsal neurovascular structures of the clitoris



UROGENITAL TRIANGLE: Male & Female External Genitalia

UG Δ: Erectile Tissues



Males:

Corpora cavernosa – paired & located laterally

a. Fused together in median plane anteriorly (free portion), except posteriorly where they separate to form the c**rura of the penis** (attached portion).
b. Crura are firmly bound to the ischiopubic ramus & adjacent perineal membrane.
c. Surrounded by the **ischiocavernosus muscles**, which contract during erection.

Corpus spongiosum – single & located ventrally

a. Surrounds the spongy urethra
b. Proximal end is bulbous, attached to the perineal membrane and known as the **bulb of the penis** (attached portion).
c. Covered by **bulbospongiosus muscles** joined at a midline raphe.
d. Free portion lies anterior to the two corpora cavernosa in the shaft

Females:

Corpora cavernosa clitoris – paired & located laterally

a. Crura are attached to the ischiopubic ramus & adjacent perineal membrane
b. Covered by **ischiocavernosus muscles**; help maintain erection
c. Form the root & body of clitoris (free portion)

Bulb of the vestibule – paired & located lateral to the vestibule

a. Attached to the perineal membrane; equivalent to the bulb of the penis b. Each bulb continues anteriorly & joins near the shaft of the clitoris.
b. Covered with ***bulbospongiosus muscle*** that helps maintain erection.

Glans Clitoris – attached to the distal end of the body

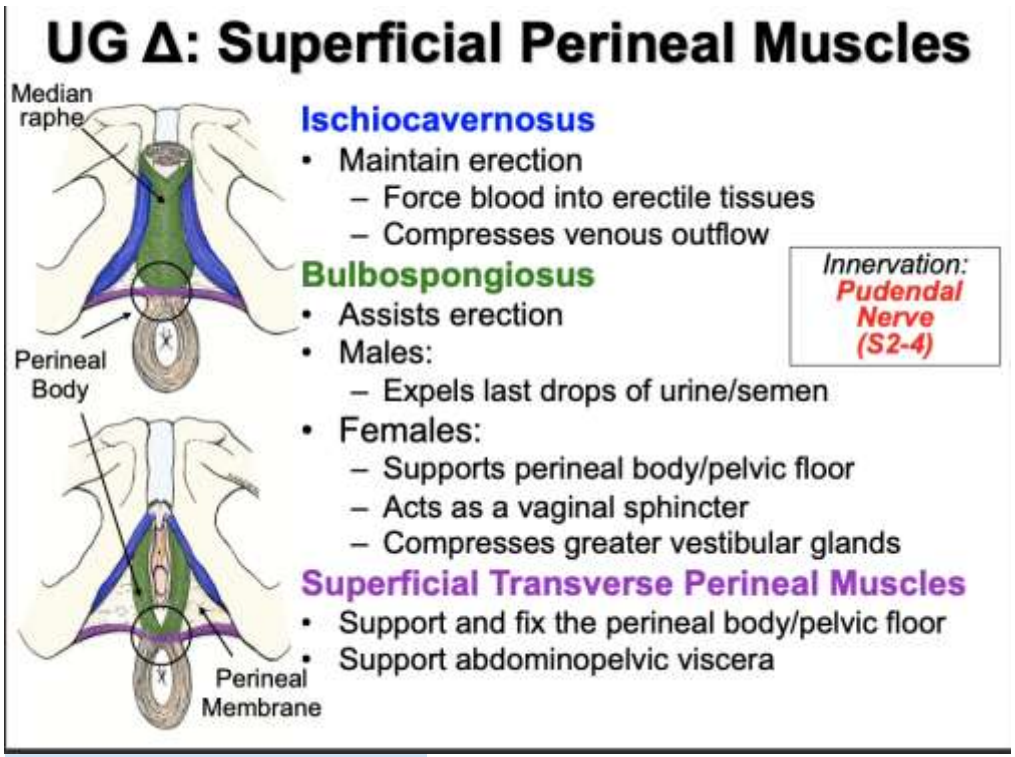
a. Is connected to the bulbs via small bands of erectile tissue (however, these bands are not considered part of the clitoris)
b. Unlike the penis, functions solely as an organ of sexual arousal

e. Distal end is expanded into the ***glans penis***, which fits closely over the blunt rounded ends of the corpora cavernosa.
   i. ***Bulbourethral Glands:*** Deep to the bulb of the penis, within the deep pouch. These glands secrete mucous into the urethra during sexual arousal to help lubricate the urethra and neutralize any remnants of acidic urine.

(highly sensitive & enlarges on tactile stimulation)

Greater Vestibular Glands (Bartholin Glands)

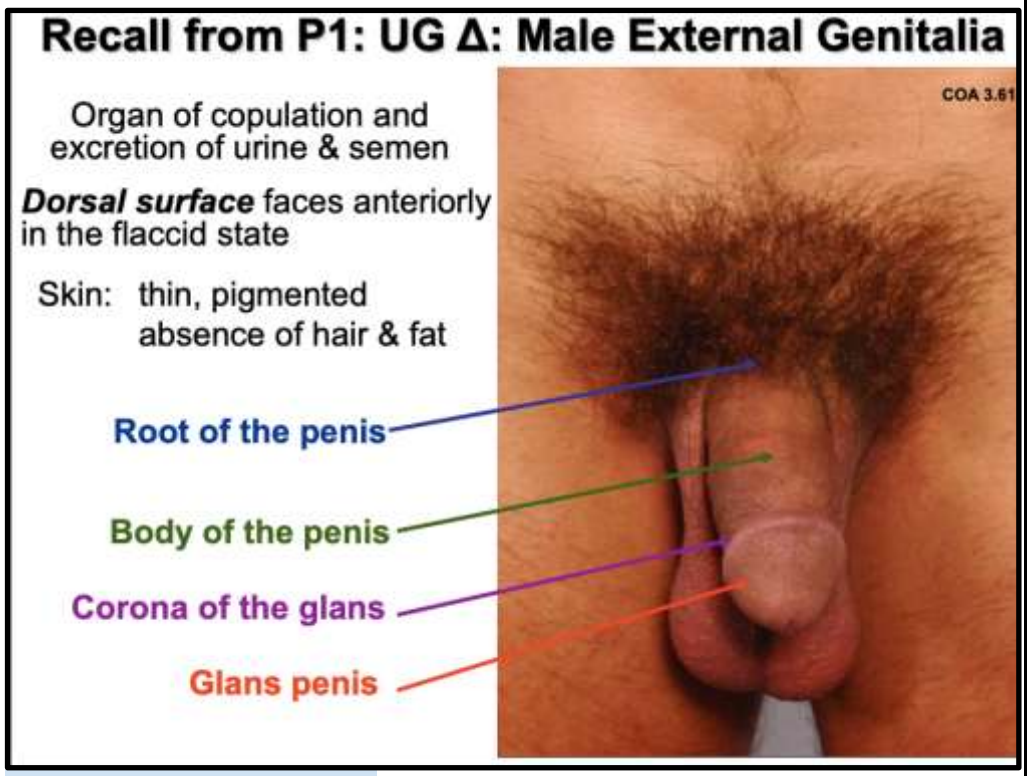
a. Located on either side of the vestibule, in the superficial perineal pouch at the posterior pole of the vestibular bulbs.
b. Secrete mucus into the vestibule during sexual arousal



Superficial Perineal Muscles (Innervation: Deep perineal nerve)

a. **Ischiocavernosus Muscles** – paired muscles covering the crura of the penis and clitoris
   i. Attach to the ischiopubic rami & perineal membrane
   ii. Maintain erection (penis or clitoris) by compressing outflow veins and
   iii. forcing blood from the root of the penis/clitoris into the body of the penis/clitoris

b. **Bulbospongiosus Muscles** – paired muscles covering the bulb of the penis or bulb of the vestibule
   i. **Males:**
      1. Attaches to the perineal body & perineal membrane; joined by a median raphe
      2. Compress the bulb of the penis to expel the last drops of urine/semen.
      3. Assists erection by compressing venous outflow (deep dorsal vein) and by increasing pressure on erectile tissue in the root of the penis.
   ii. **Females:**
      1. Attaches to the perineal body and pubic arch; passes on each side of the lower vagina, enclosing the bulb & greater vestibular glands.
      2. Assists in erection of the clitoris, compresses greater vestibular glands, & acts as a "sphincter" of the vagina
c. **Superficial transverse perineal muscles** – paired muscles that help stabilize the perineal region
   i. Originate on the ischiopubic rami & ischial tuberosity.
   ii. Pass along inferior aspect of posterior border of perineal membrane to perineal body
   iii. Support & fix the perineal body/pelvic floor to support abdominopelvic viscera



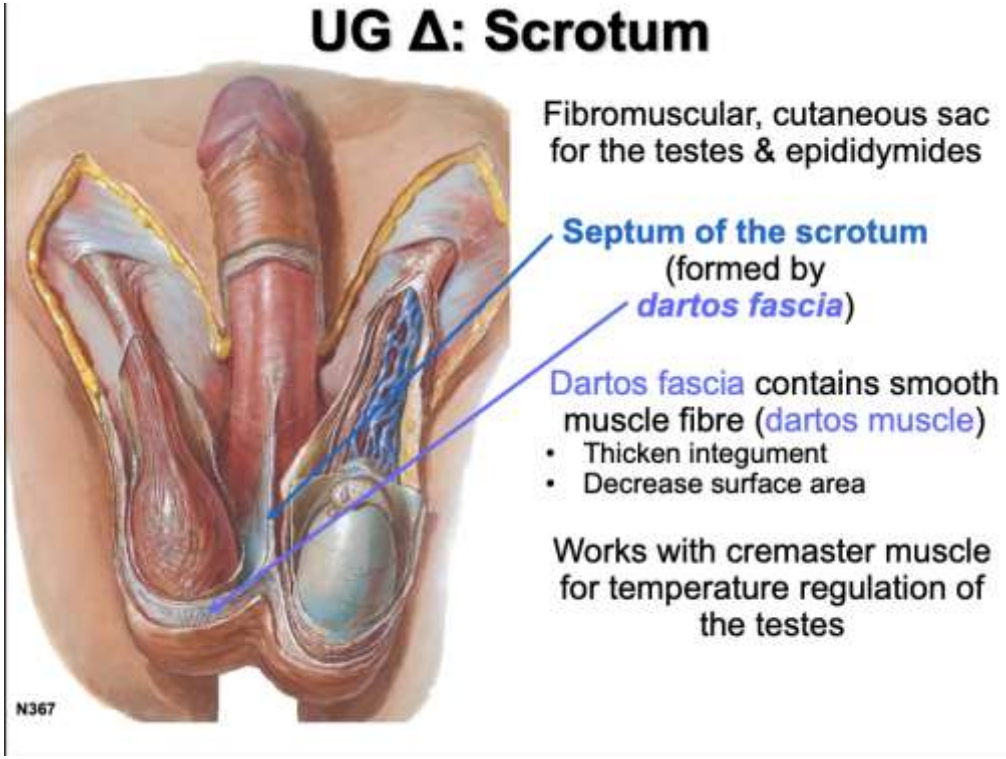
External Genitalia Males:

Penis: Organ of copulation and excretion of urine & semen

a. ***Anatomical position*** = erect penis; so when flaccid, its dorsum is directed anteriorly.
b. ***Root of the penis*** – attached portion
    i. Consists of the crura, bulb, and ischiocavernosus and bulbospongiosus muscles.
    ii. Located in the ***superficial perineal pouch***
c. **Body of the penis** – free pendulous part suspended from the pubic symphysis; does not contain any muscles
d. **Glans penis** – conical-shaped head of the penis
    i. **Corona of the glans** – margin of glans that overhangs the neck of the glans
    ii. **Neck of the glans** – grooved constriction separating the glans from body of the penis

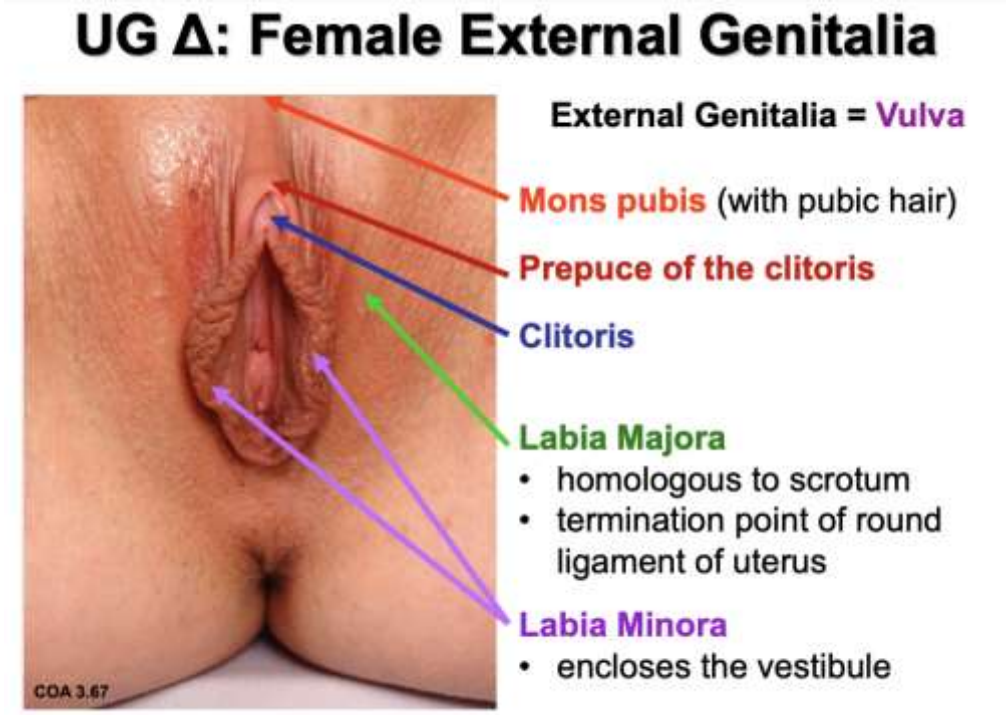
Layers:

a. **Skin** – thin, darkly pigmented
    i. Connected to the tunica albuginea, which surrounds erectile bodies, by loose connective tissue
    ii. **Prepuce** (foreskin) – double layer of skin & fascia at the neck of the glans, which in uncircumcised males covers the glans to a variable extent
    iii. **Frenulum of the prepuce** – median fold passing from the prepuce to the urethral surface of the glans
b. **Superficial fascia** – fat-free subcutaneous fascia
    i. Continuous with the scrotal dartos fascia and the superficial perineal fascia
c. **Deep fascia of the penis (Buck's fascia)** – continuation of the deep perineal fascia,
    i. Forms a strong membranous covering for the corpora cavernosa & corpus spongiosum, binding them together
    ii. **Suspensory ligament of the penis** – deep fascia arising from the anterior surface of the pubic symphysis; passes inferiorly & splits forming a sling that attaches to the penis & anchors the erectile bodies to the pubic symphysis.



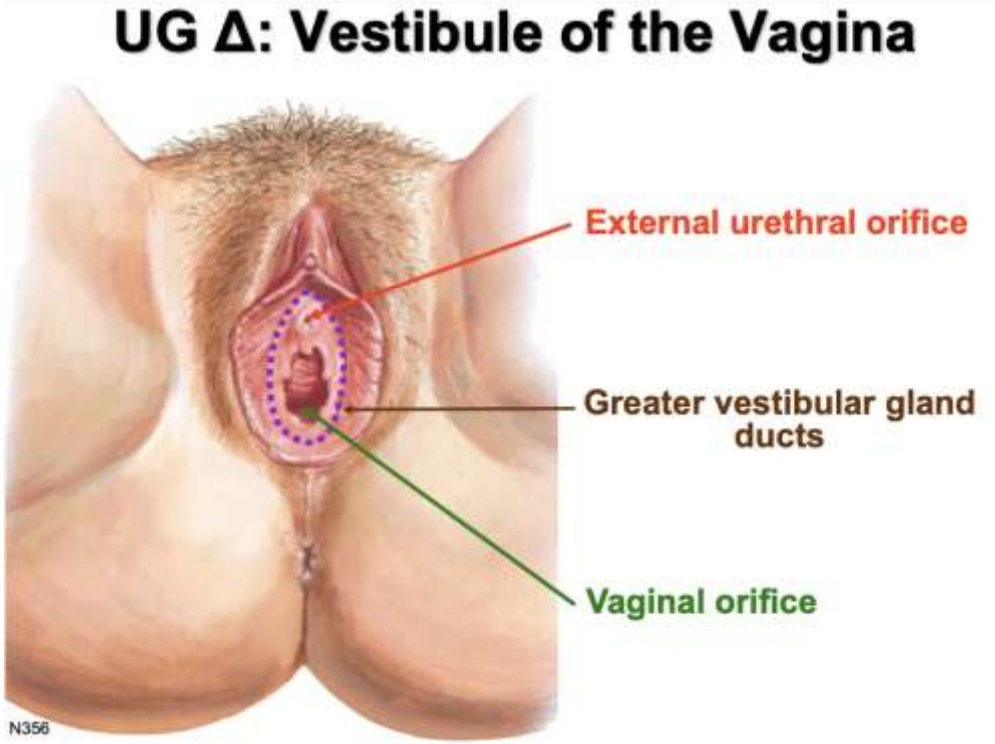
Scrotum - Fibromuscular, cutaneous sac for the testes & epididymides, located posteroinferior to the penis & inferior to the pubic symphysis

a. Scrotal raphe – the bilateral embryonic formation of the scrotum is indicated by the scrotal raphe; continuous with the **penile** and **perineal raphae.**
b. Internally (deep to the scrotal raphe), the scrotum is divided into 2 compartments by the **septum of the scrotum**, a prolongation of ***dartos fascia.***
c. **Dartos fascia** contains smooth muscle fibers and is in close contact with the skin, ***dartos muscle*** contracts the skin, decreasing surface area and increasing the integument thickness, to help with temperature regulation in conjunction with the cremaster muscle



Females: (Vulva)

a. **Mons Pubis:** Fatty eminence anterior to the pubic symphysis, pubic tubercles, & superior pubic rami; covered by pubic hair after puberty.
b. **Labia Majora** (***homologous to the scrotum***)
    i. Folds of skin that provide protection for the urethral & vaginal orifices.
    ii. Termination point of the round ligament of the uterus
    iii. Covered with pigmented skin that contains sebaceous glands & pubic hair
c. Labia Minora
    i. Rounded folds of fat-free, hairless skin that enclose the vestibule of the vagina
    ii. The lateral aspect of the labia minora unite anterior to the glans of the clitoris forming the **prepuce** (foreskin or hood) **of the clitoris.**

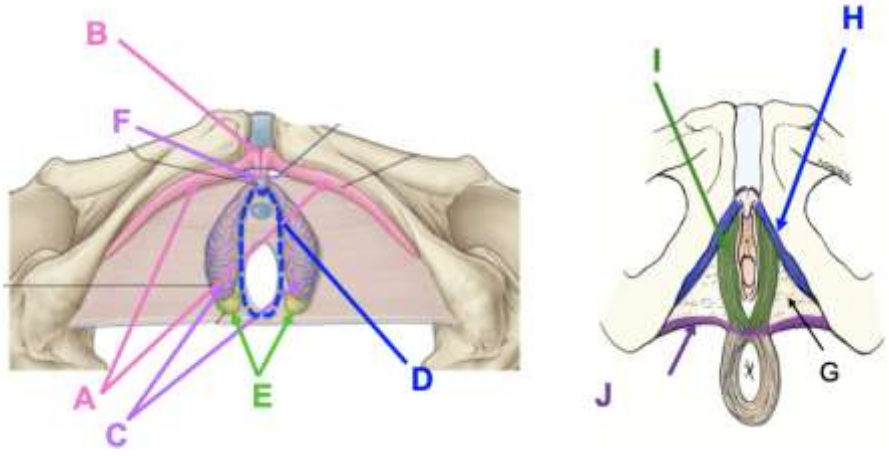


Vestibule of the vagina: space surrounded by the labia minora

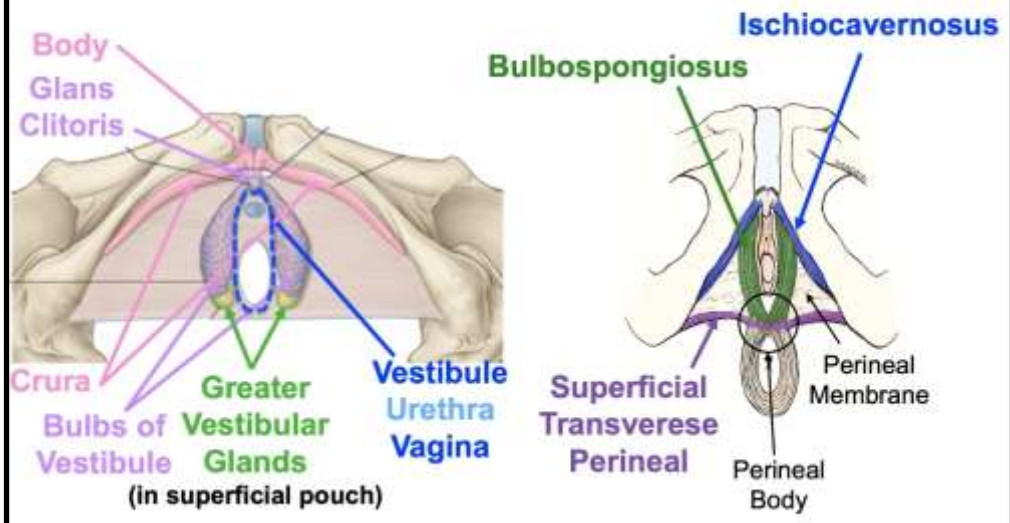
Contents:

a. **External urethral orifice:** Located ~2 cm posterior & inferior to the glans.
   i. **Clinical Note:** This is a tightly closed opening. When inserting a catheter into the bladder, the catheter tip will tend to enter the larger nearby opening (the vagina). If this happens, the tip becomes contaminated with vaginal flora. The non-sterile catheter must be discarded and a fresh package opened or you will give the patient a UTI (urinary tract infection).
   ii. **Vaginal orifice:**
      1. Size & appearance vary with the condition of the hymen, a thin, annular fold of mucus membrane immediately within the vaginal orifice surrounding the lumen.
      2. ***Greater vestibular glands:*** located on the upper, medial aspects of the labia minora.
         a. Secrete mucous into the vestibule during sexual arousal.

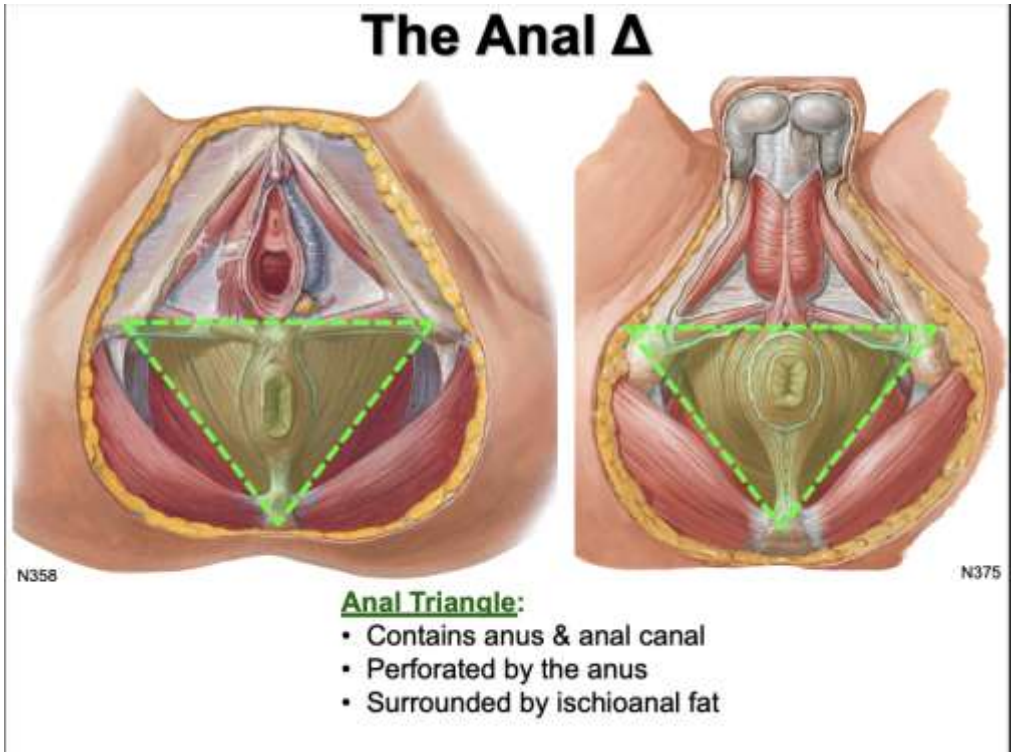
Knowledge Check

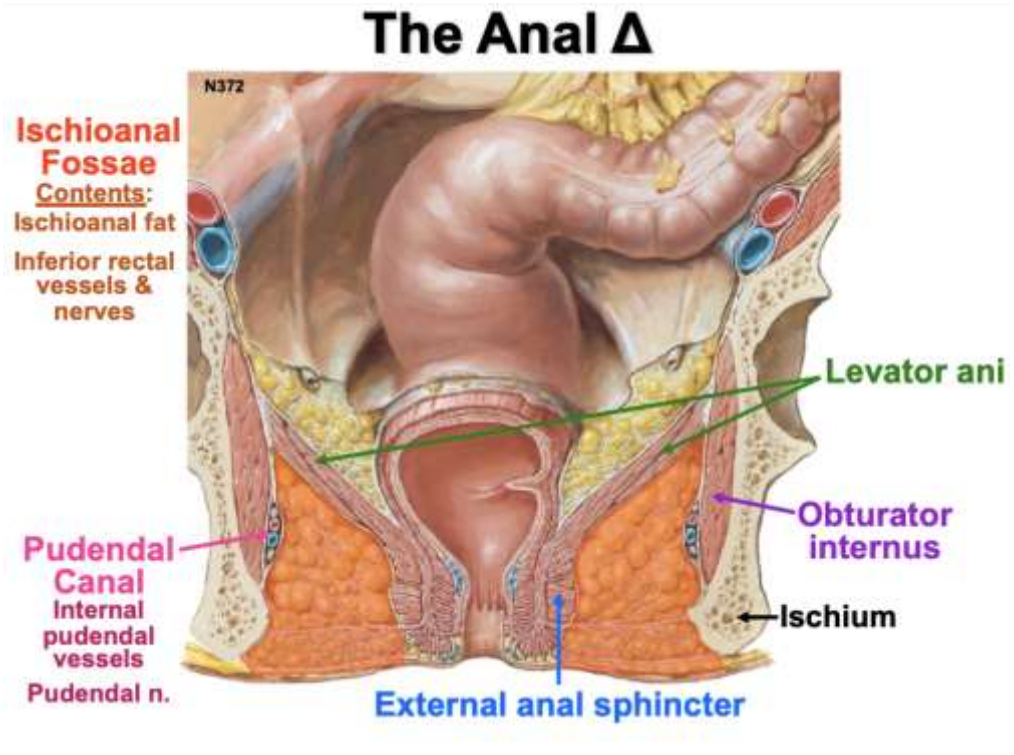


Knowledge Check Feedback



ANAL TRIANGLE

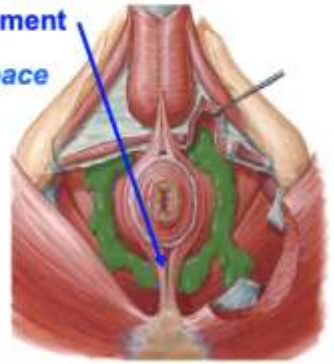




Anal Triangle: Contains the anus, anal canal, & associated musculature and is surrounded by ischioanal fat.

a. **Ischioanal Fossae**
   i. Wedge-shaped spaces on either side of the anal canal between the skin of the anal region & pelvic diaphragm
   ii. **Borders:**
      1. ***Anteriorly:*** bodies of the pubic bones inferior to the origin of puborectalis
      2. ***Laterally:*** ischium & inferior part of obturator internus muscle & fascia
      3. ***Medially:*** sloping border of the pelvic diaphragm & external anal sphincter
      4. ***Posteriorly:*** sacrotuberous ligament & gluteus maximus
   iii. Occupied by ***ischioanal fat*** & loose connective tissue
      1. Readily displaced to permit descent & expansion of the anal canal during defecation or vagina during childbirth.
      2. Traversed by the inferior rectal vessels & nerves.
   iv. The two ischioanal fossae communicate via the deep postanal space between the anal canal and the tip of the coccyx, deep to the anococcygeal ligament
      1. **Clinical Note:** Ischioanal fat is among the last reserves of fatty tissue to disappear during starvation. In the absence of support provided by the ischioanal fat, rectal prolapse is relatively common.

Anal Δ: Ischioanal Abscess



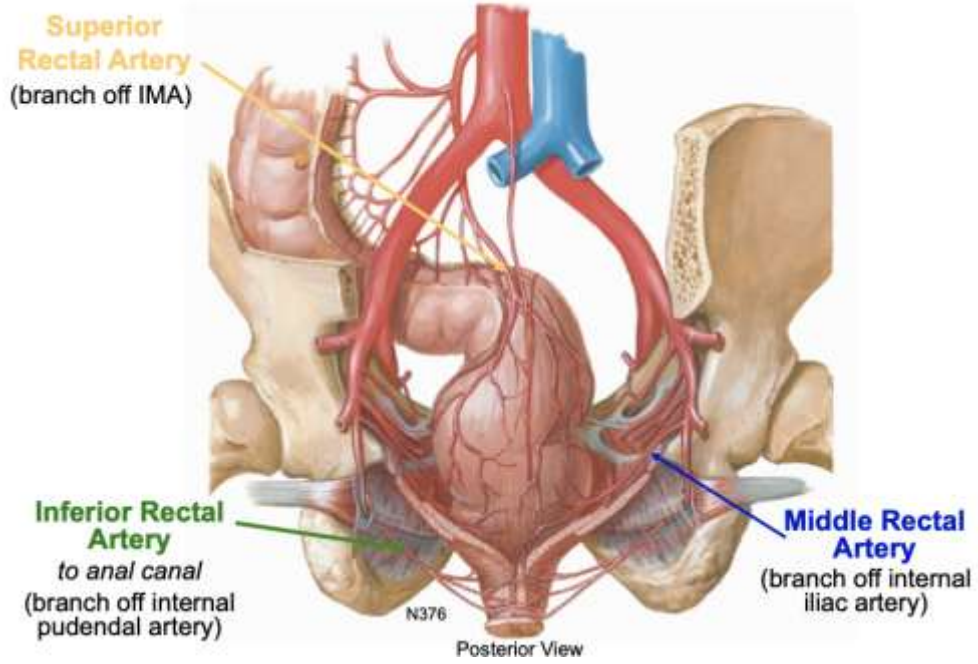
- Painful infections in the ischioanal fossae
  - Puss can be palpated as fullness & tenderness
- Most commonly caused by inflammation of anal glands
- Ischioanal fossae communicate posteriorly
  - Infection can spread to other side forming a **"horseshoe-shaped" abscess**
- Treated via surgical incision and drainage



Clinical Note: Ischioanal abscesses are painful infections found in the ischioanal fossae.

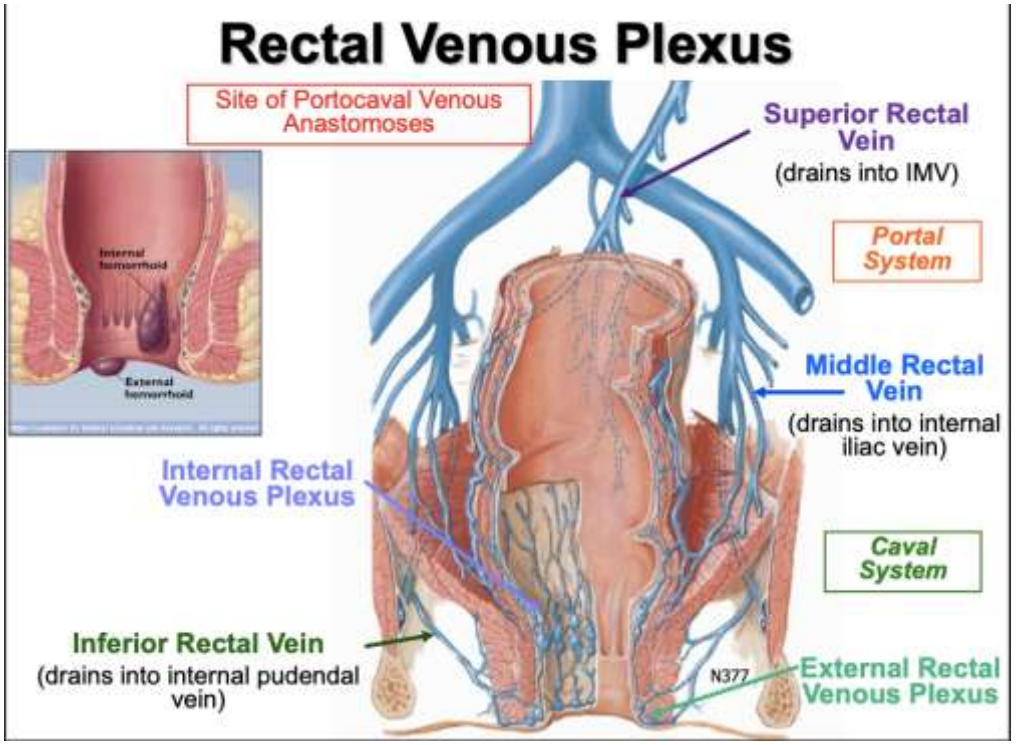
a. Pus can collect and be palpated as fullness & tenderness between the anus & ischial tuberosity.
b. The infection may reach the ischioanal fossae after cryptitis (inflammation of anal sinuses), extension from pelvirectal abscesses, after a tear in the anal mucous membrane, or from a penetrating wound in the anal region.
c. Since the ischioanal fossae communicate posteriorly through the deep postanal space, an abscess in one fossa may spread to the other side forming a semicircular "**horseshoe-shaped" abscess** around the posterior aspect of the anal canal.

Arterial Anastomoses Around Rectum



Pudendal Canal (of Alcock)

a. Horizontal passageway in the lateral wall of the ischioanal fossa that runs within the obturator fascia on the medial aspect of obturator internus
b. Gives passage to the **internal pudendal vessels** & p**udendal nerve**
   i. Proximally, branches to give rise to the ***inferior rectal vessels***, which supply the *external anal sphincter & perianal skin*
   ii. Distally, bifurcates to give rise to the ***Dorsal artery & nerve of the penis/clitoris*** – run in the deep perineal pouch



Rectal Venous Plexus

a. Venous drainage will drain in a similar pattern to arterial drainage
b. Separated into two parts:
   i. **Internal Rectal Venous Plexus**
      1. Above the pectinate line
   ii. **External Rectal Venous Plexus**
      1. Below the pectinate line
c. Venous Plexuses are ***3 main veins***:
   i. Superior rectal Vein → inferior mesenteric vein → **PORTAL SYSTEM**
   ii. Middle Rectal Vein → internal iliac vein → **CAVAL SYSTEM**
   iii. Inferior Rectal Vein → internal pudendal vein → **CAVAL SYSTEM**

Rectal Venous plexus = site of portacaval venous anastomoses (varicose veins within this region = hemorrhoids)

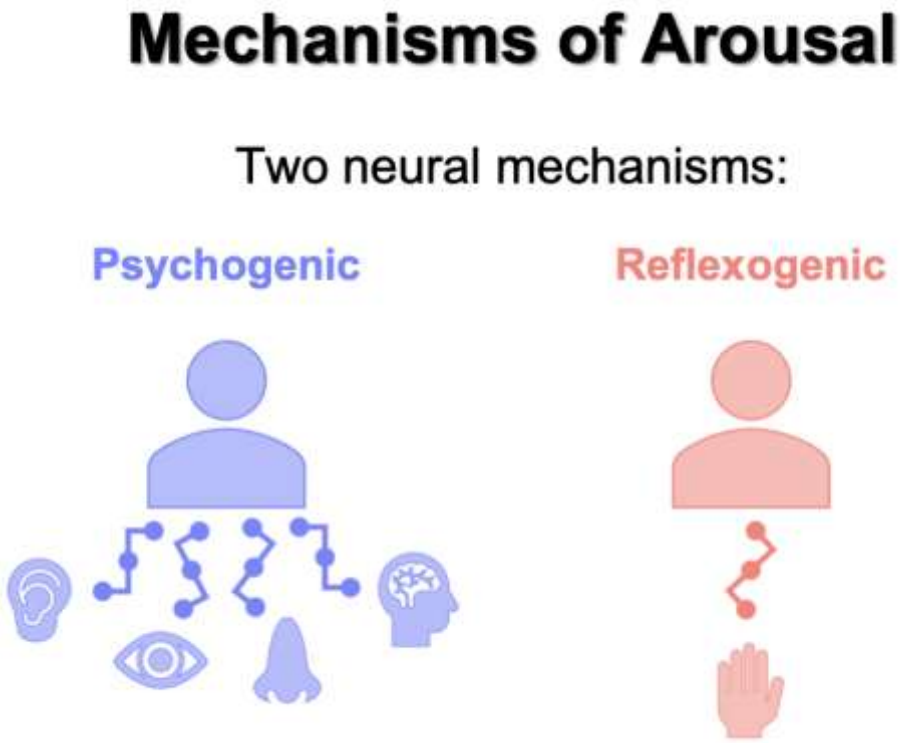
- Happens when this area is subject to high pressure (eg. pregnancy)
- Anal canal divided by the pectinate line into :
  - **Upper 2/3 portion** derived from hindgut endoderm → innervated by autonomic pelvic splanchnic
  - **Inferior ⅓ portion** derived from ectoderm → somatically innervated by the pudendal nerve (nerve to the perineum)
    - **Clinically relevant** because it determines how patients experience pain or not with hemorrhoids

The anal triangle is the posterior triangle of the diamond shaped perineum. It contains:

- The terminal part of the digestive system (anal canal)
- Ischioanal fat
- Pudendal Canal

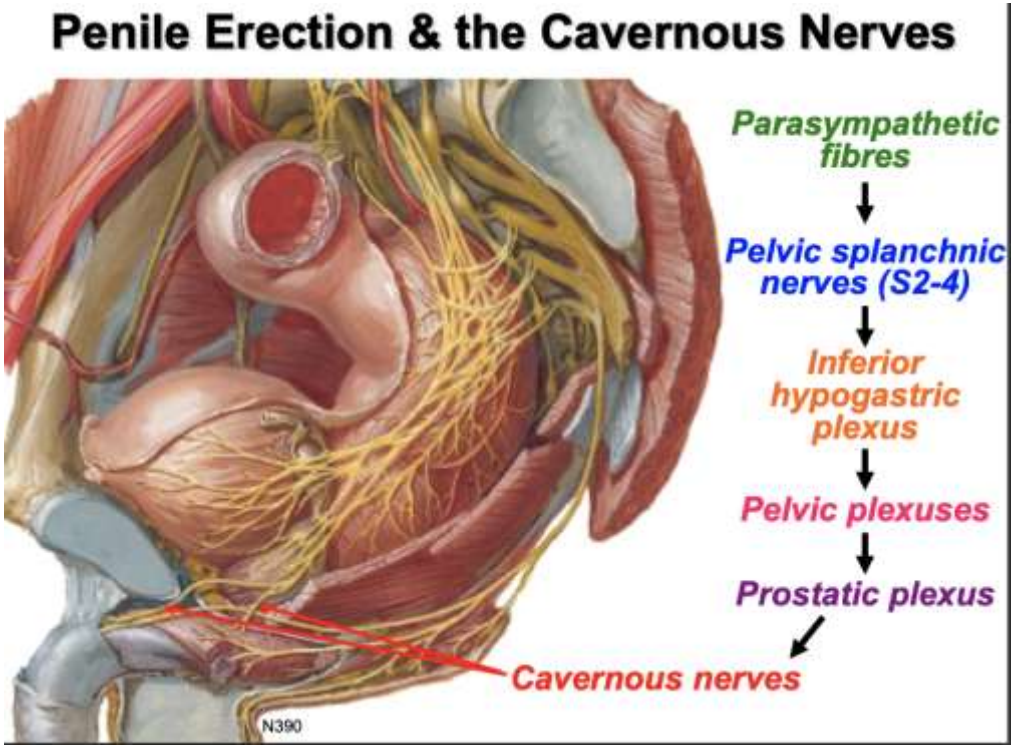
The anal canal is a site of portocaval venous anastomosis. If these veins become varicosed they become hemmorhoids. Internal hemmorhoids are not painful because that part of the anal canal is under autonomic control. While external hemorrhoids are painful due to the somatic innervation of the inferior anal canal.

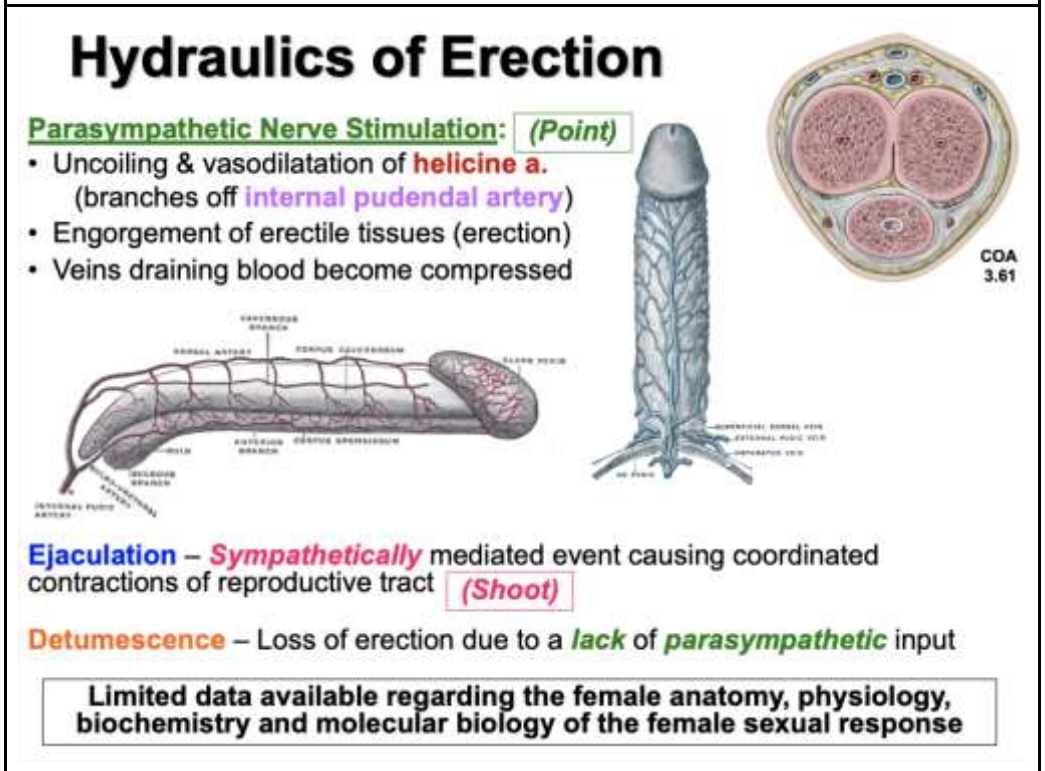
FUNCTIONAL SEXUAL ANATOMY



Functional Sexual Anatomy:

a. There are two neural mechanisms recognized in humans: psychogenic and reflexogenic
   i. **Psychogenic erections** take place in response to auditory, visual, olfactory, or imaginative stimuli, for example, and involve pathways from the brain to the sacral segments of the spinal cord, where preganglionic cells of the pelvic splanchnic nerves are stimulated. (There is inconclusive evidence that there might be a sympathetic component to psychogenic erections).
   ii. **Reflexogenic erections** involve tactile stimulation, for example, of the penis. Afferent fibres in the pathway travel primarily in the dorsal nerve of the penis, back to the sacral part of the spinal cord, where the parasympathetic efferent pathway is stimulated.





Hydraulics of Penile Erection:

a. ***Parasympathetic fibers*** destined for the genital erectile tissues leave the CNS via ventral rami of spinal nerves *S2-4* as **pelvic splanchnic nerves.**
b. Pelvic splanchnic nerves merge with *inferior hypogastric nerves* to form *pelvic plexuses.*
c. Fibers destined for the erectile tissue travel to subdivisions of the pelvic plexus, which surround the capsule around the prostate gland (***prostatic plexus***).
d. These **cavernous nerves** directly penetrate the perineal membrane to reach erectile tissues.
    i. **Clinical Note:** Prostatic removal from an outside approach is likely to interfere with this route. Hence loss of erection is a frequent outcome of radical prostatectomy.
e. Cavernous bodies consist internally of a sponge-like meshwork of endothelial- lined spaces known as ***sinuses*** that become engorged with blood. Other than during an erection, most blood is shunted away from the erectile sinuses toward the deep dorsal vein of the penis through arteriovenous anastomoses. Mental sexual excitement or physical stimulation of the somatic afferent fibers from the external genitalia or the internal filling of bladder or rectum initiate **parasympathetic stimulation** (via cavernous nerves) of the:
    a. Arteriovenous anastomoses (nearly closing them, blocking outflow route)
    b. **Helicine (spiral) arteries** within the erectile tissue "uncoil" & dilate resulting in engorgement of erectile tissues (erection).

As expansion occurs, the thin-walled veins draining the corpora become compressed, so more blood flows into the corpora than venous blood outflows. An erection is maintained until orgasm occurs or the erotic stimulus passes.

a. **Ejaculation** is a ***sympathetically*** mediated event causing coordinated contractions throughout the male reproductive

POINT (parasympathetic) & SHOOT (sympathetic)

tract to result in ejaculation.

b. ***Detumescence*** – stage following ejaculation or loss of erection. Due to the lack of parasympathetic input and is a reversal of the hydraulic events described above.

Erectile Dysfunction

The inability to obtain an erection (impotence)

- Causes:
  - CNS or endocrine
  - Lack of nervous stimulation
  - Lack of blood vessel dilation
    - Treated with oral medication
  - Prostatic plexus or cavernous nerve lesion
    - Surgical Treatment
      - Penile Pump
      - Semi-rigid Rods









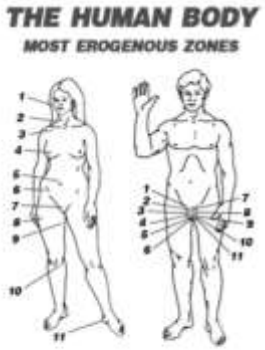
Clinical Note: Erectile Dysfunction is the inability to obtain an erection (***impotence***) and has many possible causes. Central nervous system disorders (***hypothalamic***) or endocrine disorders (***pituitary or testicular***) may result in reduced testosterone secretion impeding possible erection. Nerve fibres may fail to stimulate erectile tissues, or blood vessels may be insufficiently responsive to autonomic stimulation preventing erection. In these cases erection can be achieved with the assistance of oral medications that increase blood flow to erectile tissue. If there is a lesion of the prostatic plexus or cavernous nerves causing impotence a surgically implanted, semi-rigid or inflatable, penile prosthesis may assume the role of the erectile bodies.

a. The mechanism (anatomy, physiology, biochemistry, and molecular biology) of the female sexual response is not fully understood and limited data are available. Thus, large gaps exist in our knowledge of how the CNS controls female sexual function.

Sexual Response Cycle

Sexual Response (physiological and psychological):

1. **Desire** – Want/need to partake in sexual activity induced by external or internal stimuli
2. **Arousal** – a state with feelings and physiologic changes involving the genitals
   a. Central mechanisms – Thoughts, dreams, fantasies
   b. Non-genital peripheral mechanisms – salivation, sweating, cutaneous vasodilation and nipple erection
   c. Genital mechanisms – clitoral, labial, and vaginal engorgement, penile erection
3. **Orgasm** – altered state of consciousness associated with general sensory input, male ejaculation
4. **Resolution**



Sexual Response Cycle

a. Sexual response has both physiological and psychological responses
   i. **Desire** – Want/need to partake in sexual activity induced by external or internal stimuli
   ii. **Arousal** – a state with feelings and physiologic changes involving the genitals
      1. **Central mechanisms** – Thoughts, dreams, fantasies
      2. **Non-genital peripheral mechanisms** – salivation, sweating, cutaneous vasodilation and nipple erection
      3. **Genital mechanisms** – clitoral, labial, and vaginal engorgement, penile erection
      4. **Orgasm** – altered state of consciousness associated with general sensory input, male ejaculation
      5. **Resolution**

ⓘ The sexual response cycle is a complex physiological and psychological response.

- Psychogenic Arousal may have many sources of sensory and imaginary input
- Reflexogenic Arousal is somatic sensory controlled
- Erection is parasympathetically controlled
- Ejaculation is sympathetically controlled